MILITARY SPECIFICATION

REQUIREMENTS FOR TRAINING SYSTEM DEVELOPMENT

This specification is approved for use by the Naval Training Equipment Center (NAVTRAEQUIPCEN), Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the conduct of instructional systems development (ISD) activities involved in the design, production, and implementation of materials for operator, maintenance, and instructor training courses (ITC). The ISD procedures are applicable to both existing and emerging weapon systems. Specifically, it addresses the procedures to be used in accomplishing the following ISD steps:

(a) Problem analysis (PA).

(b) Training development and support planning analysis.

(c) Task listings development.

(d) Entry level behavior analysis.

(e) Objectives hierarchies development.

(f) Training equipment survey.

(g) Media selection.

(h) Syllabi development.

(i) Training support requirements analysis (TSRA).

(j) Lesson specification development.

(k) Lesson format guide development.

(l) Lesson materials and test development.

(m) Quality control (QC) planning.

(n) Implementation planning.

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(o) Instructional system operation.

(p) Training device operation.

(q) Student orientation.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for the bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-847	Format Requirements for Scientific and Technical Reports prepared by or for the Department of Defense.

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Training system PA. For an existing weapon system, a PA shall be performed to assess the characteristics and resources of an operator or maintenance training program and to determine deficiencies, training problems, and training program needs. The analysis shall provide the data necessary to develop a valid training development and support plan (TDSP) which will form the basis for effective project procurement and management and will provide input into the training program development process.

3.1.1 Methodology. The PA shall consist of a set of data collection and assessment activities performed for the purpose of establishing operator or maintenance program training needs. It shall provide a detailed, systematic, and comprehensive accounting and description of all critical training program requirements, training resources, constraints, and deficiencies. The PA procedures shall include, but not be limited to, those discussed in the following paragraphs.

3.1.1.1 Initial guidance meeting. After consultation and coordination with the cognizant Navy organizations, the contractor shall convene a meeting of appropriate representatives. Attendees shall be advised to come prepared with names, building, and phone numbers of data collection sources and points of contact who are knowledgeable in the following areas:

(a) Learning centers.

(b) Media library/storage.

(c) Training facilities.

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(d) Academic training.

(e) Operational training or maintenance practical job training (PJT) departments.

(f) Media production facilities (unit, base, naval district, etc.).

(g) Support facilities (ranges, gaming areas, etc.).

(h) Other.

Representatives from the following commands shall attend:

(a) NAVTRAEQUIPCEN.

(b) Type commanders (PAC and LANT).

(c) Cognizant training activity.

(d) Parent command of (c).

(e) Cognizant training support activities.

(f) Operational unit (if specified by procuring activity).

(g) Other representatives specified by the procuring activity.

The purpose of the meeting shall be to clearly set forth the purpose, scope, plan, and schedule of the pending PA, to include funding and time constraints and specific data collection/coordination roles and responsibilities. The meeting shall also address the status of any training program enhancement efforts, such as new trainer procurements, new facilities (learning centers or ranges, etc.), and any previous decisions, plans, or pending contracts. The meeting attendees shall be sufficient to surface and discuss all significant previous or pending decisions and how they would or should relate to any future training program enhancement project.

3.1.1.2 On-site survey and analysis. The contractor shall visit, survey, and analyze the applicable training system(s) and associated command(s) for the purpose of identifying specific project requirements, capabilities, facility and command project support readiness, and anticipated project problem areas. By means of direct interviews with cognizant Navy personnel at each visited site, reviews of existing technical publications and other relevant data, and by physical inspection of training materials, facilities, and processes, the contractor shall perform the following tasks:

(a) Survey chain of command. Consider all direct or supporting elements of the chain of command, with particular emphasis on those elements through which organizational access will be required throughout training program planning and development.

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(b) Survey existing training program. Consider purpose and scope of program, program management, instructors, syllabi, media, student evaluation techniques, student flow rate, training schedule, training equipment, aircraft, facilities, program support factors, and weapon system life cycle.

(c) Identify anticipated training program changes. Consider new mission requirements, new tactics, weapon system changes, revised training facilities and equipment, revised student flow rate, and revised training schedules.

(d) Identify actual or potential problems, deficiencies, and unavailable or unsuitable project resources. Consider problems and deficiencies resulting from analyses of information collected in (b) and (c) above and resource constraints which could affect performance of an ISD or other training enhancement program.

(e) Make recommendations. Consider all data collected and analyses performed to recommend primary and alternate approaches to training program enhancement (e.g., complete ISD, reduced ISD, upgrade of existing materials, etc.) and include supporting rationale.

(f) Estimate costs. Consider all primary and alternate approaches identified in (e) above and break down costs into categories to enable comparisons among approaches.

3.1.1.3 PA review meeting. After the PA has been completed, a meeting shall be convened to present the results of the PA and to plan for the TDSP. Attendees shall include representatives of the commands identified in 3.1.1.1 and others as appropriate. Revisions/recommendations agreed upon shall be included in the problem analysis report (PAR).

3.1.2 PAR. Procedures used during the PA, data analyses, and recommendations shall be documented in accordance with UDI-H-25710 and submitted to the contracting agency for review.

3.2 TDSP. A TDSP shall be developed prior to the initiation of a training program development project. The TDSP shall define participant roles and responsibilities and the approach, methods, and procedures for resolving the problems identified in the PAR. Variables, such as project objectives, phasing, procurement strategy, internal Navy coordination, required resources, and others shall be considered so as to arrive at a method for achieving project goals (both technical and managerial) that is objective and cost effective for the government. The output of the TDSP shall be a well-defined, comprehensive, and fully coordinated program for upgrading the existing training program.

3.2.1 Methodology. TDSP development, at a minimum, shall be conducted in accordance with the processes described in the following paragraphs.

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3.2.1.1 Review of the PAR. To ensure that the TDSP is directed toward documented, relevant problems, development shall be initiated by a thorough review of the PAR. This review shall focus on program objectives, training assets and constraints, alternative recommendations, training activity and support commands, and others as appropriate.

3.2.1.2 Plan development. Information contained in the PAR and decisions resulting from the coordination meeting shall be used to develop the TDSP. Development of a TDSP is a creative process which will differ from project to project. At a minimum, topics discussed in the following paragraphs shall be considered.

3.2.1.2.1 Project objectives. Technical and managerial objectives to be achieved during the anticipated training program development and scope of the program (e.g., maintenance training, "O" level only) shall be determined.

3.2.1.2.2 Roles and responsibilities. Organizations participating in the management of, or contributing to, the program shall be identified and the specific role, responsibility, and function clearly delineated. These roles and responsibilities should be displayed in a functional matrix type chart which includes the organization and title and/or title and code number of the office or individual responsible. Official addresses, names of individuals, and their telephone numbers (points of contact) should be included. Special management or technical teams, such as fleet teams, should be identified in a separate functional matrix chart. Specific resource commitments required of participating organizations will be identified.

3.2.1.2.3 Program phasing and procurement strategy. Optimal phasing and procurement strategies to meet project objectives shall be determined. Phasing shall be either single or multiple (e.g., a complete program versus analysis phase, design phase, production phase). Selection criteria shall be cost effectiveness, quality, minimal delays, and advantages of different types of procurement practices for different phases. Additional coordination meetings shall be held as required to select the optimal phasing and procurement strategy from options developed by the contracting agency. The selected approach shall provide the greatest advantage to the government.

3.2.1.2.4 Coordination. For each phase identified above, all tasks required to accomplish the technical and managerial objectives shall be specified. Tasks in each phase of an ISD program shall be accomplished as described in 3.4 through 3.28 of this specification. For each task in each phase, one or more responsible organizations and coordination among all involved organizations shall be identified. The specification of roles and responsibilities shall be consistent with existing instructions and policies. The coordination and communication plans shall address responsibility for convening meetings, dissemination of technical and managerial data, internal organizational coordination requirements, as appropriate, the methodology for program status/progress reporting, and the frequency for periodic meetings and reports. Specific data requirements, when known, shall be included.

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3.2.1.2.5 Schedules and milestones. Critical milestones (i.e., phases and tasks) and a schedule for completion of the milestones consistent with any time demands identified in the PAR shall be developed. The schedule shall be presented in both an overall phasing schedule and in individual task schedules within phases. Milestones shall be clearly indicated. The schedule shall provide the best estimate, the most pessimistic estimate, and the most optimistic estimate of duration.

3.2.1.2.6 Resource estimates. All resources required to carry out the program objectives, beyond those resources (assets) on hand, shall be identified. These resources shall include, but not be limited to:

(a) Contractor manpower (technical and support for the different phases and tasks).

(b) Navy manpower (subject matter experts [SMEs], support, etc.).

(c) Facilities (work space, specialized facilities, etc.) and materials (paper, tapes, etc.).

(d) Equipment (media, reproduction, word processors, etc.).

(e) Training equipment modification.

(f) Technical publications.

(g) Replacement parts and spares provisioning.

Once identified, the resource requirements shall be costed using a contractor-supplied cost analysis model and current cost information available from NAVTRAEQUIPCEN data base. To aid in budgeting, a best estimate, an optimistic estimate, and a pessimistic estimate shall be provided. The difference between the optimistic and pessimistic estimates shall be determined by such factors as government SME availability, contractor costs, facilities availability, and equipment purchase or modifications.

3.2.1.2.7 Special planning requirements. Special requirements resulting from the PAR, coordination meetings, and the TDSP development process which require separate recognition and planning shall be identified. These requirements will be project-specific and, therefore, cannot be defined in this specification. Examples of special requirements are as follows:

(a) Computerized management system.

(b) Adoption of pending policy changes.

(c) Anticipated engineering change procedures (ECPs).

3.2.1.3 TDSP review meeting. After the draft TDSP has been developed, a meeting shall be held to review the TDSP, assign roles and responsibilities, and obtain resource commitments. All organizations to be involved in carrying out or supporting the training program shall be represented.

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3.2.2 TDSP report. The TDSP shall be prepared in accordance with UDI-H-25711 and submitted to the contracting agency for review.

3.3 TDSP update. Revisions to the TDSP shall be made as required by the contracting agency before and during training program development. Input information shall be furnished by the contracting agency, the contractor, and other relevant organizations/individuals.

3.3.1 Methodology. TDSP updates shall be conducted as specified in 3.2.1. Coordination meetings shall be convened as required and all affected topic areas shall be updated to reflect program changes.

3.3.2 TDSP update report. The updated TDSP shall be prepared in accordance with UDI-H-25711 and submitted to the contracting agency for review.

3.4 Operator task listings. Task listings containing all operator mission tasks shall be developed and validated. The validated task listings shall provide the basis for development of the training system.

3.4.1 Operator task listings development. The output of operator task listing development shall be a comprehensive set of task statements exhaustively defining all types of missions, mission phases, and major job tasks for each crew position. The task listing shall include task statements reflecting both normal and emergency crew activities. Each task statement shall be presented in the general form of a behavioral objective with the action specified, along with the conditions of performance.

3.4.1.1 Data sources. Sources for development of operator task listing data include, but are not limited to:

(a) Mission analysis data.

(b) Functions analysis data.

(c) Human factors engineering task analysis data.

(d) SMEs.

(e) Design engineering data.

(f) Similar existing systems' task listings.

(g) Previous task listings for the system under study.

3.4.1.2 Methodology. Operator task listings development for each job shall be conducted in accordance with the process shown in Figure 1 and as described in the following paragraphs. The relationship existing between various levels in the task listing, with examples, is shown in Figure 2.

3.4.1.2.1 Step 1: List major missions within a job. The following criteria shall be considered when determining major missions. A mission:

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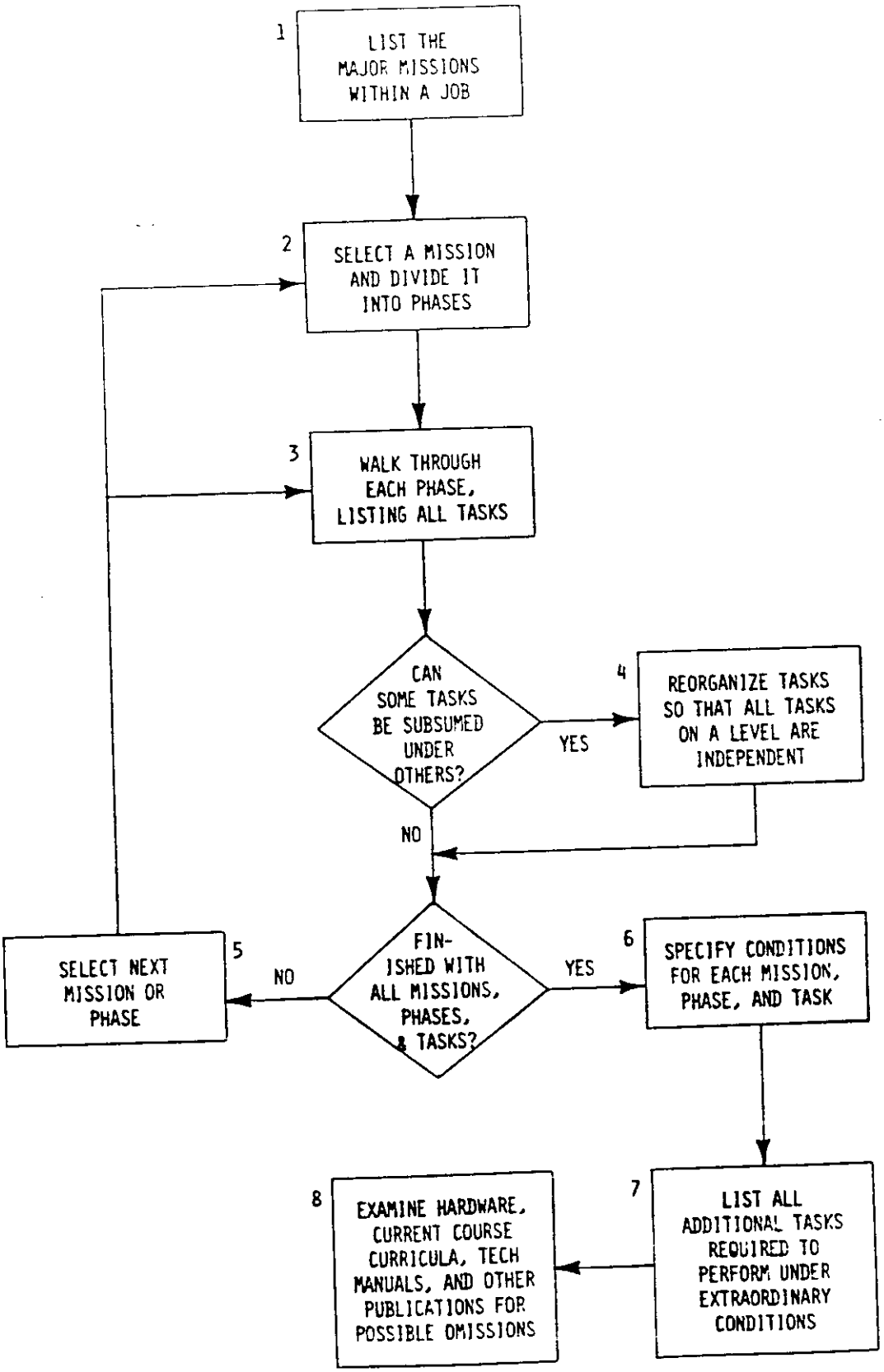


FIGURE 1. Operator task listing process.

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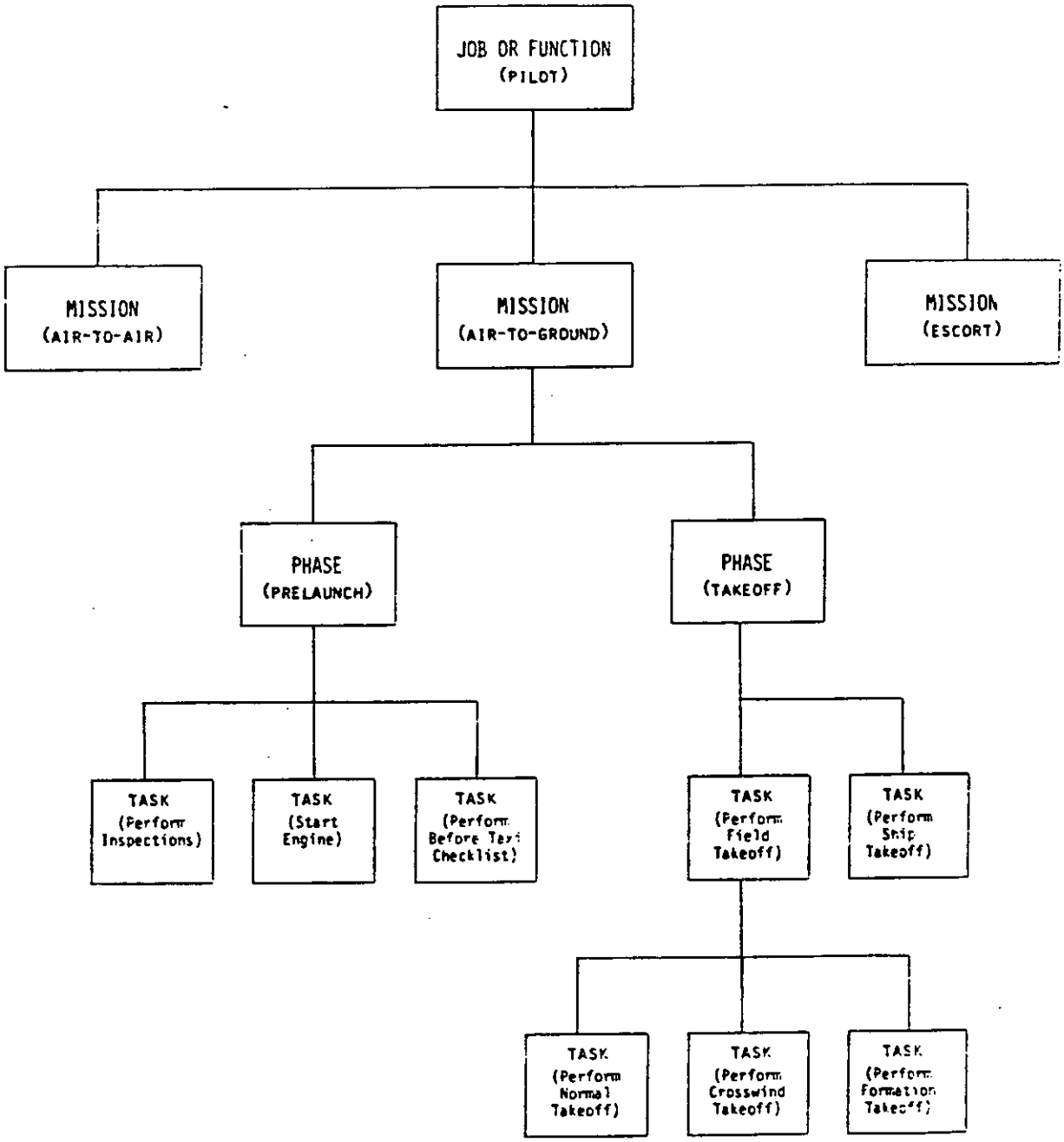


FIGURE 2. Overall structure of a task listing.

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(a) Has a set of activities usually performed by a system of individuals.

(b) Has a clearly definable beginning and end point. Not all task listings will have multiple distinct missions.

(c) Is often identified with an end goal of coordinated crew activity.

3.4.1.2.2 Step 2: Select a mission and divide it into phases. The following characteristics shall be considered when identifying mission phases. A mission phase has the following characteristics:

(a) It can be given a name.

(b) It has a logical beginning and end point.

(c) It occupies an exclusive time slice.

(d) All phases taken together describe the entire mission.

3.4.1.2.3 Step 3: Walk through each phase, listing all tasks. Tasks are names for the products they create or the processes they use. Phases are named for the time periods they occupy. The two can be distinguished on this basis. The following characteristics shall be considered when identifying tasks:

(a) It is a significant operator activity (with a name).

(b) It has an observable beginning and end point or results in a consistent product.

(c) It usually includes a consistent sequence of specific behaviors (sometimes called "subtasks").

In developing task listings, sufficient detail shall be provided to enable accurate validation of the listing. Without sufficient detail, important tasks may be omitted from training or be assigned to an incorrect level/location of training. On the other hand, detail which appropriately should be added during the development of objectives hierarchies should not be included. This will encumber the validation and task selection process. Generally, the appropriate level of detail can be determined as follows:

(a) The point beyond which task components, rather than whole tasks, will be entered.

(b) The lowest level at which performance will be evaluated independent of other contiguous tasks.

Figure 3 contains a portion of a task listing with comments indicating the appropriateness of each entry.

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	TASK	COMMENT
1.6.3	Deliver ORD	Insufficient detail because there are multiple methods.
1.6.3.1	Deliver ORD by dive delivery	Insufficient detail because there are multiple methods.
1.6.3.1.1	Deliver ORD by low angle dive delivery	Proper level of detail.
1.6.3.1.1.1	Configure A/C for low angle dive delivery	Too much detail. This is clearly a task component.
1.6.3.1.2	Deliver ORD by high angle dive delivery	Proper level of detail.
1.6.3.1.2.1	Configure A/C for high angle dive delivery	Too much detail.
1.6.3.2	Deliver ORD by LABs delivery	Insufficient detail because there are multiple methods.
1.6.3.2.1	Deliver ORD by loft delivery.	Proper level of detail.

FIGURE 3. Sample task listing indicating different levels of detail.

3.4.1.2.4 Step 4: Organize tasks so that all tasks on a level are independent. Each task shall be reviewed by asking the question, "Can this task be subsumed under another task?" If it can, then it shall be deleted from its existing position and put on a lower level. The task statements at each level in the analysis shall be independent of each other - neither subordinate nor superordinate. The task listing shall form the upper levels of a hierarchy. The lower levels will be added when objectives hierarchies are developed (3.8).

3.4.1.2.5 Step 5: Select next mission or phase. The task listing process shall be continued until all missions and phases have been analyzed.

3.4.1.2.6 Step 6: Specify conditions for each mission, phase, and task. Conditions are the "givens" of a performance. They describe the circumstances under which the task is performed. Conditions shall include, but not be limited to, the following:

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(a) Environmental factors (sensory conditions, such as space, light, quiet or noise, temperature, weather, wind, pressure, system conditions).

(b) Other people factors (alone, part of a team, under supervision . . .).

(c) Equipment factors (what job aids, tools, or equipment are available . . .).

(d) Information (what relevant information is available; checklists, charts . . .).

(e) Problem definition (what stimulus is present to initiate the task; system characteristics that provide cues and "feel").

(f) Time (task duration, pacing).

(g) Concurrent tasks.

3.4.1.2.7 Step 7: List all additional tasks required to perform under extraordinary conditions. This step is accomplished by asking, for each mission, phase, and task identified, "Are there any conditions under which this is performed which require deviations from the normal procedures?" Extraordinary conditions create new tasks previously overlooked in the task listing. Any new tasks identified by analysis of extraordinary conditions shall be incorporated in the task listing and treated as all other tasks.

3.4.1.2.8 Step 8: Re-examine all data sources (e.g., hardware, current curricula, tech manuals, and other publications) to identify possible omissions in the task listing. At a minimum, mission analysis, function analysis, and human factors task analysis reports shall be examined.

3.4.1.3 SME training for operator task listings development. In order for SMEs to function adequately during the development of operator task listings, training shall be provided in the following areas:

(a) Familiarization with task listing process.

(b) Identification of, and differentiations between, major missions, phases, and tasks.

(c) Development of instructional objectives.

(d) Use of standardized numbering and reference system.

(e) Use of data sources.

(f) QC and review procedures.

(g) Application of the task listing to subsequent ISD steps.

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Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.4.2 Operator task listings validation and task selection. The output of the operator task validation and task selection step of ISD shall be a comprehensive list of tasks to be trained. The tasks included in the task listings shall be validated and then classified in terms of training requirements.

3.4.2.1 Methodology. For existing weapons systems, validation information shall be collected from personnel experienced in operation of the weapon system. These shall include operational and training SMEs, and others as deemed appropriate. For emerging weapon systems, validation information shall be collected from personnel experienced in similar, existing weapon systems and from engineers and other personnel involved in designing, developing, and testing the weapon system. The contractor shall provide task listing validation and selection procedures for approval by the contracting agency. At a minimum, task list validation and task selection shall be conducted as described in the following paragraphs.

3.4.2.1.1 Review of similar task listings. Task listings from similar weapons systems and any previous task listings from the system under study shall be reviewed and compared with the task listing being validated. This will help ensure accuracy and completeness. Modifications shall be made as required.

3.4.2.1.2 Task listings validation and task selection conference. A conference shall be convened to obtain concurrence and sign-off on the tasks to be included in the ISD training curriculum. Attendees for existing and emerging weapon systems shall be as outlined in 3.4.2.1. A representative cross-section of qualified personnel must attend the conference. The output of the task listings validation and selection conference shall be a set of tasks which all parties at the conference concur should form the basis for the training program. Each task selected for training shall be classified in one of the following categories:

(a) Deferred training: Tasks which require training facilities, personnel, equipment, or time beyond that available in the immediate training activity and which could be successfully trained at a later time.

(b) Full-scale training: Tasks which are critical, are frequently performed, are above entry level, and are not feasible for deferred training.

(c) Review only training: Tasks which are critical or frequently performed and have been previously learned, but about which there is doubt concerning full retention of what was learned.

(d) No training: Tasks which are judged to be noncritical, trivial, or rarely performed in the mission environment and tasks

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which are entry level behaviors of incoming trainees. These tasks shall not be included in the training program, but shall be marked and maintained on the task list in the event mission requirements or entry level skills change.

3.4.2.2 SME training for operator task listings validation and task selection. In order for SMEs and other personnel as appropriate (e.g., design engineers, test pilots, etc.) to properly validate the task listing and select tasks for training, instruction shall be provided in the following areas:

(a) Structure of the task listing.

(b) Overview of task listing development procedures.

(c) Procedures for reviewing similar or previous task listings.

(d) Procedures for conducting the task listing validation and task selection conference.

(e) Guidelines for differentiating among types of training for tasks (i.e., no training, review, full, deferred).

(f) Application of the task listing to subsequent ISD steps.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.4.3 Operator task listings report. Procedures used during task listing development, task listing organization and format, validation and task selection procedures, and a validated task listing for each crew position (including category of training for each task) shall be documented in accordance with UDI-H-25713 and submitted to the contracting agency for review.

3.5 Maintenance task listings. Task listings containing all maintenance tasks shall be developed and validated. The validated task listings shall provide the basis for development of the training system.

3.5.1 Maintenance task listings development. The output of maintenance task listings development shall be a comprehensive set of task statements exhaustively defining all of the responsibility areas, systems, subsystems, and major job tasks related to each maintenance position. The task listings shall include task statements reflecting both normal and emergency activities. Each task statement shall be presented in the general form of a behavioral objective with the action specified, along with the conditions of performance.

3.5.1.1 Data sources. Sources for development of maintenance task listing data include, but are not limited to:

(a) Phase maintenance requirements cards for each job category.

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(b) Maintenance instructional manuals (MIMs).

(c) Manufacturers' maintenance instructions.

(d) SMEs rated in appropriate job categories (i.e., power plants, hydraulics, radar, electronics, etc.) and level of maintenance (i.e., organization, intermediate, PAR).

(e) Previous task listings on the same/similar systems and subsystems.

3.5.1.2 Methodology. Maintenance task listings shall be conducted in accordance with a process to be developed and provided for approval by the contracting agency. One acceptable method is provided in 6.3.2 of this specification. Consideration shall be given to listing those tasks that are necessary to training as distinguished from operational tasks.

3.5.1.3 SME training for maintenance task listings development. In order for SMEs to function adequately during the development of maintenance task listings, training shall be provided in the following areas:

(a) Familiarization with task listing process.

(b) Identification of, and differentiations between, major responsibility areas, systems, subsystems, and tasks.

(c) Development of instructional objectives.

(d) Use of standardized numbering and referencing system.

(e) Use of data sources.

(f) QC and review procedures.

(g) Application of the task listing to subsequent ISD steps.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.5.2 Maintenance task listings validation and task selection. The output of the maintenance task validation and task selection step of ISD shall be a comprehensive list of tasks to be trained. The tasks included in the task listings shall be validated and then classified in terms of training requirements.

3.5.2.1 Methodology. For existing weapon systems, validation information shall be collected from personnel rated in each job category for which the maintenance task listing was developed. For emerging weapon systems, validation information shall be collected from personnel in each job category applicable to maintenance of the emerging system and from engineers and other personnel involved in designing, developing, and testing the system.

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The contractor shall provide maintenance task listing validation and selection procedures for approval by the contracting agency. At a minimum, task list validation and task selection shall be conducted as described in the following paragraphs.

3.5.2.1.1 Review of similar task listings. Task listings from weapons systems having the same or similar aircraft systems and subsystems and any previous task listings from the system under study shall be reviewed and compared with the task listing being validated. This will help ensure accuracy and completeness. Modifications shall be made as required.

3.5.2.1.2 Review by job category. SMEs in each job category (i.e., power plants, hydraulics, radar, electronics, etc.) and each level of maintenance (i.e., organization, intermediate, PAR) shall review that portion of the task listing relevant to their job category and level. Reviewers shall be drawn from operational and training communities and be graduates of schools commensurate with their positions (i.e., A school for organizational, B school for intermediate, etc.). The output of each set of job category by level reviews shall be a set of tasks which all reviewers concur should form the basis of the training program. Each task selected for training shall be classified in accordance with (a) through (d) in 3.4.2.1.2 of this specification. In addition, tasks for which job aids are necessary or desirable shall be identified.

3.5.2.2 SME training for maintenance task listings validation and task selection. In order for SMEs in the different job categories and levels and other personnel as appropriate (e.g., design engineers, reliability and maintainability engineers) to properly validate the task listing and select tasks for training, instruction shall be provided in the following areas:

(a) Structure of the task listing.

(b) Overview of the task listing development procedures.

(c) Procedures for reviewing similar or previous task listings.

(d) Procedures for SME review by job category.

(e) Guidelines for differentiating among types of training for tasks (i.e., no training, review, full, deferred).

(f) Guidelines for identifying tasks for which training aids are necessary or desirable.

(g) Application of the task listing to subsequent ISD steps.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.5.3 Maintenance task listings report. Procedures used during task listing development, task listing organization and format, validation and

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task selection procedures, and a validated task listing for each job category and level (including category of training for each task and training aid requirements) shall be documented in accordance with UDI-H-25713 and submitted to the contracting agency for review.

3.6 Operator and maintenance task listings update. The task listings shall be updated on a periodic basis to insure that the most current data possible is available for use in subsequent ISD steps.

3.6.1 Data sources. The most recent information available shall include, but not be limited to, the areas listed below to identify possible additions to, or modifications of, the task listings:

(a) Mission analysis data.

(b) System test data.

(c) Functions analysis data.

(d) Fleet maintenance data.

(e) SMEs.

(f) Equipment ECPs.

(g) Human factors engineering task analysis data.

(h) Phase maintenance requirements cards.

(i) MIMs.

(j) Manufacturers' maintenance instructions.

3.6.2 Methodology. Once it is determined that a change must be made to a portion of a task listing, the affected tasks shall be deleted or modified as necessary and the revised tasks shall be inserted into the listing using the appropriate reference numbering. If additional tasks are needed, the numbering of other tasks shall be modified to allow for new task insertion.

3.6.3 Task listings update report. Task listing updates for each crew position (operator) or job category and level (maintenance) shall be documented as required in accordance with UDI-H-25713 and submitted to the contracting agency for review.

3.7 Analysis of entry level behavior. An analysis of entry level behavior shall be conducted as part of the ISD for an emerging weapons system. The output of the analysis of entry level behavior shall be a comprehensive description of trainee skills and knowledge upon entry into training. For an existing weapons system, the analysis of entry level behavior shall not be required, but rather shall be an integral component of the task listing, validation, and selection process and the development of objectives

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hierarchies, in that the processes and products of these steps will reflect levels of entry behavior as determined by the developers.

3.7.1 Methodology. The student entry level analysis for an emerging weapons system shall be performed to characterize the student population entering the training program. Using data provided by the government concerning potential sources of students and expected levels of training to be accomplished and data collected during the PA, a profile of the "average" and range of entry level behavior to be expected shall be determined. The analysis shall be keyed to the tasks requirements and systems components of the emerging weapons system. The profile of entry level behavior shall be used to assist in the development and validation of task listings, the selection of tasks for training, and as applicable in subsequent ISD steps.

3.7.2 SME training for entry level behavior analysis. Since entry level behavior analysis is a contractor activity, no formal SME training shall be required. SME input shall be solicited as required during the analysis process.

3.7.3 Student entry level report. Entry level behavior shall be documented in accordance with UDI-H-25714 and submitted to the contracting agency for review.

3.8 Objectives hierarchies. Those tasks which were included in the training program as a result of the task validation and selection process shall be analyzed in further detail during objectives hierarchy development by instructional developers (IDs) working in conjunction with SMEs. The output of this step of the ISD process shall be a comprehensive set of behavioral objectives and hierarchy diagrams specifying the component subtasks, skills, and knowledges for every task in the task listing. The objective shall describe the action to be performed, the conditions for performance (i.e., job aids, equipment, tactical information, etc.) and performance measures (i.e., speed, accuracy, etc.). The hierarchy diagrams shall specify the subordinate, superordinate, and independent relationships between objectives.

3.8.1 Data sources. Sources of data for the development of objectives hierarchies include, but are not limited to:

(a) Mission analysis data.

(b) Functions analysis data.

(c) Human factors engineering task analysis data.

(d) SMEs.

(e) Phase maintenance requirements cards.

(f) MIMs.

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(g) Manufacturers' maintenance instructions.

(h) Design engineering data.

(i) Similar existing systems' objectives hierarchies.

(j) Previous objectives hierarchies for the system under study.

3.8.2 Methodology. Objectives hierarchy development shall be conducted in accordance with the process shown in Figure 4 and as described in the following paragraphs.

3.8.2.1 Step 1: List major tasks. This step shall be accomplished during the task listing step of ISD. It is included here to emphasize that the task list is the input to the objectives hierarchy development procedure.

3.8.2.2 Step 2: Select the next major task for analysis. The following considerations shall be given to the selection of the next major task:

(a) Select a task which involves a large part of the job time.

(b) Select a task which requires many sub-procedures for its completion.

(c) Select a task which tends to be one of the most difficult.

3.8.2.3 Step 3: Divide the major task into component subtasks. The IDs and the SMEs shall analyze the performance components of the selected major task by identifying each of the steps or procedures required. ("Steps" or "procedures" may be decisions and judgments, as well as overt, observable acts.) These decisions and judgments shall be identified as subtasks.

3.8.2.4 Step 4: Determine whether all necessary decisions have been identified as task components. When analyzing a task that requires performance of a large number of procedures and interaction with complex system hardware, identification of each operator/maintenance personnel decision shall be made to determine whether the operator/maintenance personnel must:

(a) Determine when to perform a procedure.

(b) Determine which of several rules or procedures to use.

(c) Evaluate the adequacy of a procedure.

(d) Determine when a procedure should be stopped.

3.8.2.5 Step 5: Add decision tasks to the component list. The decision tasks identified shall be added to the component list. Task statements

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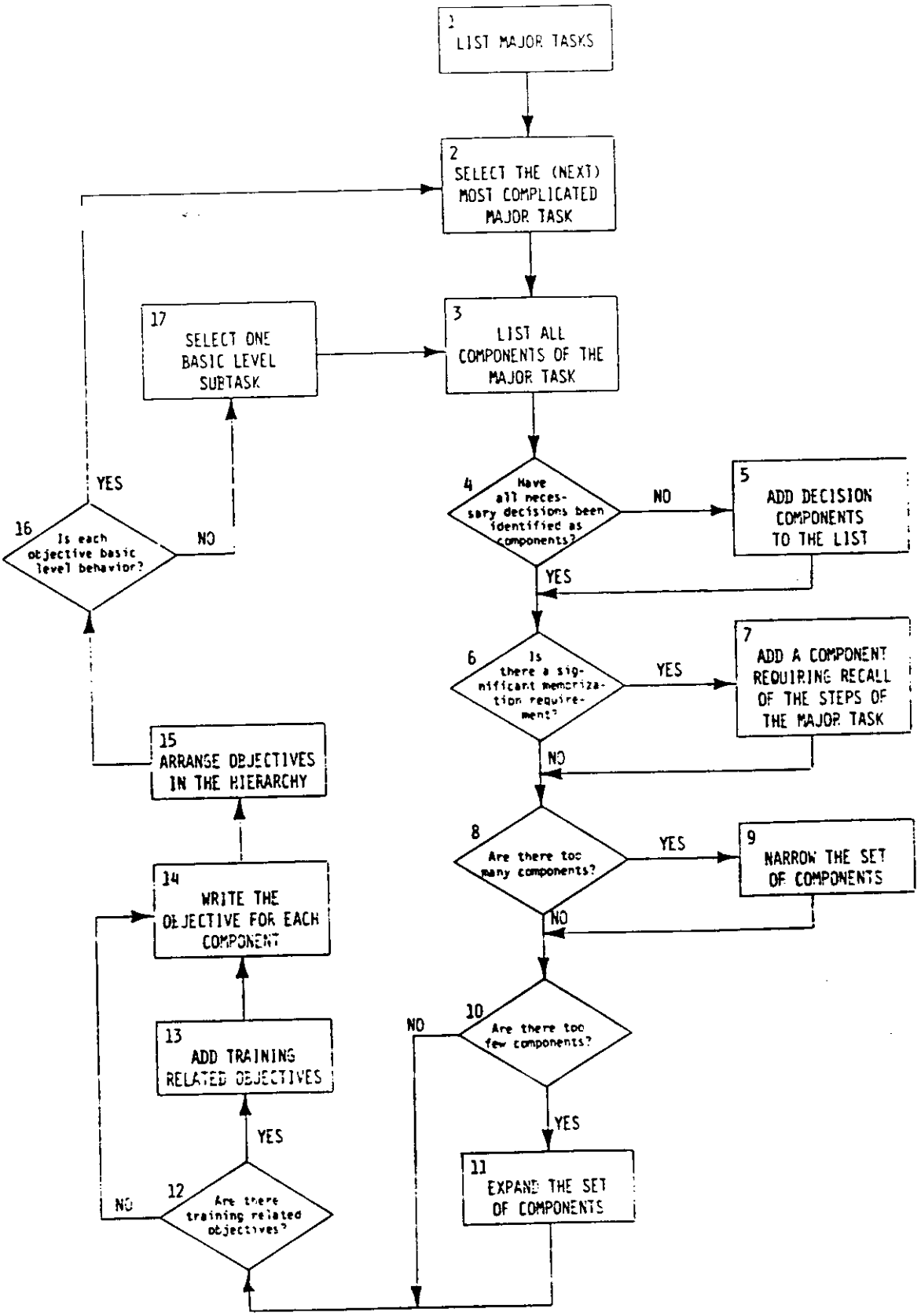


FIGURE 4. Objectives hierarchy development process.

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shall be developed to spell out exactly what decisions the operator/maintenance personnel must make in order to perform the task in all situations.

3.8.2.6 Step 6: Determine whether memorization is a significant element of the task. The memorization component of a task shall be considered significant if it is judged likely that an average trainee would be unable to perform the task as a whole because he could not remember which components must be performed or the order of their performance. If, however, a checklist is provided as a job aid for performing a high memory load task, the task shall not be identified as requiring a significant memorization component. If it is determined that the memorization requirement for the task is significant, proceed to step 7; otherwise, proceed to step 8.

3.8.2.7 Step 7: Add an objective requiring recall of the steps of the major task. The task shall be defined in terms of the steps to be memorized and the required order of performance, if order is significant. An objective for memorization of the task components shall be added at the same hierarchy level as the components.

3.8.2.8 Step 8: Determine if there are too many subtasks. The ID and SME shall determine whether any of the following criteria applies to the component list:

(a) Any subtask is a lower level component of another subtask on the list.

(b) Any subtask repeats another subtask on the list.

(c) Any subtask is not necessary to the accomplishment of the main task.

(d) Any subtask is trivial.

If it is determined that there are too many subtasks, proceed to step 9; otherwise, go to step 10.

3.8.2.9 Step 9: Narrow the list to the minimum set of subtasks necessary to perform the terminal objective. IDs and SMEs shall narrow the subtask list in one or more of the following ways:

(a) Eliminate any overlapping subtasks.

(b) Eliminate any task that is part of another subtask on the list.

(c) Eliminate subtasks that are not essential to performance of the superordinate task.

(d) Group trivial subtasks into major logical categories and designate each category as a single subtask.

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3.8.2.10 Step 10: Determine whether there are too few subtasks. If, after having mastered all given subtasks, the trainee would be unable to perform the main task without more than a few simple instructions and some coordination practice, one or more subtasks have been omitted. This analysis shall form the basis of the instructional program. A subtask left out at this point will result in instruction left out later. If Figure 5 is true, the developers shall omit step 11 and proceed directly to step 12. If the statement is not true, additional subtasks must be defined. The developers shall perform step 11 before proceeding to subsequent steps.

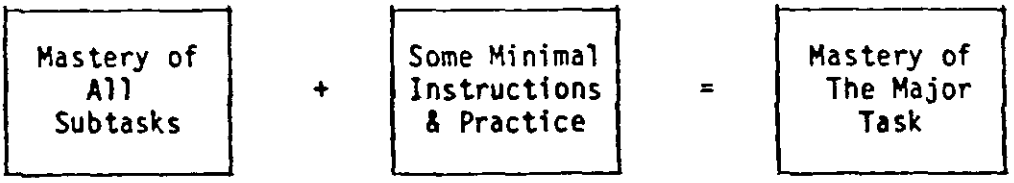


FIGURE 5. Algorithm for hierarchy completion.

3.8.2.11 Step 11: Expand the list of subtasks to the minimum necessary set. The ID shall define additional subtasks as necessary to comply with the statement illustrated in Figure 5.

3.8.2.12 Step 12: Determine if there are training related objectives. Training related objectives are those objectives that must be performed in the training environment only, as distinguished from mission objectives. They are included in the training curriculum to facilitate learning of mission objectives. Examples are touch-and-go landings and stalls. If training related objectives are identified, proceed to step 13; otherwise, proceed directly to step 14.

3.8.2.13 Step 13: Add training related objectives to the component list. The training related objectives identified shall be added to the component list. Training related objectives shall be identified through an appropriate coding procedure.

3.8.2.14 Step 14: Write an instructional objective for each component. Conditions and standards statements shall be added to the subtask component statements in order to make them complete instructional objectives. Conditions and standards should reflect real world requirements.

(a) Conditions shall be developed in accordance with 3.4.1.2.6.

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(b) Standards are the variables which will be used to evaluate performance. (Note: Standards at this step in the ISD process are measurement variables, not numerical values of variables.) Standards are not required for objectives which clearly require academic instruction only. Detailed standards will be formulated during lesson specification development. For performance objectives, standards will include, but not be limited to, the following:

(1) Time or speed.

(2) Accuracy/error rate variables (e.g., speed, heading, mechanical tolerance, etc.).

(3) Safety considerations.

(4) Process specifications (e.g., sequence of performance).

(5) Product specifications.

3.8.2.15 Step 15: Arrange the objectives in the hierarchy. The ID and SME shall arrange the instructional objectives under the superordinate objective that is being analyzed, as illustrated in Figure 6.

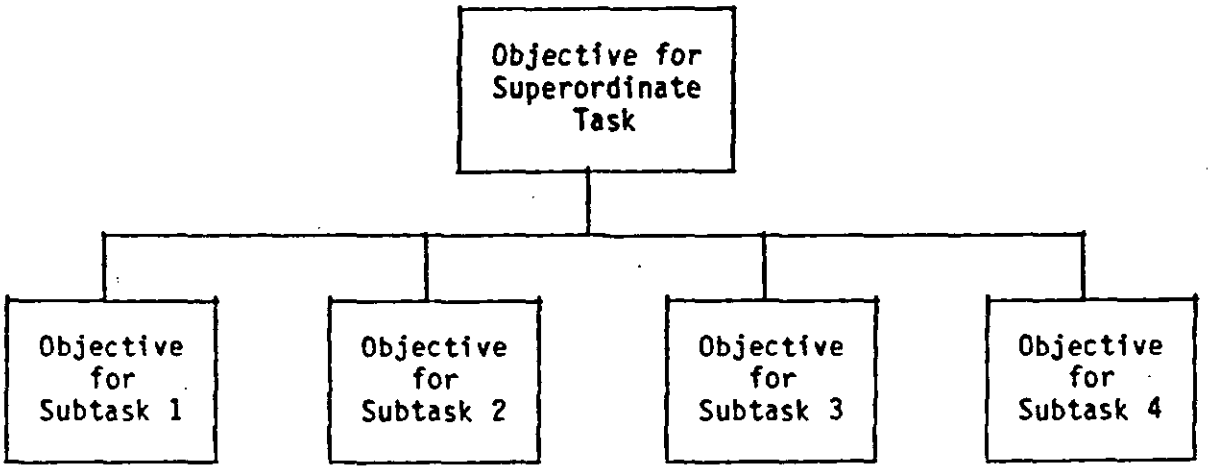


FIGURE 6. Requisite subtask hierarchy.

3.8.2.16 Step 16: Determine whether each objective is a basic level behavior. An objective shall be considered to be a basic level behavior if

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it is the lowest level behavior that is significant for training (i.e., not trivial and above entry level).

3.8.2.17 Step 17: Iterate analytical process. The ID shall select another non-basic level objective and analyze it, utilizing the same procedure followed for the other tasks (steps 3 through 16 above). This process shall be continued until all tasks have been analyzed to their basic level objectives.

3.8.3 SME training for objectives hierarchies development. In order for SMEs to function adequately during hierarchy development, training shall be provided in the following areas:

(a) Use of task listing in hierarchy development.

(b) Familiarization with the hierarchy development process.

(c) Identification of subtasks.

(d) Identification of decision component subtasks.

(e) Identification of significant memorization requirements.

(f) Identification of training related objectives.

(g) Development of instructional objectives.

(h) Use of proper hierarchy layout conventions.

(i) Use of standardized hierarchy numbering conventions.

(j) Use of objectives hierarchies in subsequent ISD steps.

(k) QC and review procedures.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.8.4 Objectives hierarchies report. Procedures used during the development of objectives hierarchies and a complete set of hierarchies for each crew position (operator) or each job category and level (maintenance) shall be documented in accordance with UDI-H-25715 and submitted to the contracting agency for review.

3.9 Objectives hierarchies update. The objectives hierarchies shall be updated at periodic intervals to insure that the most current data possible is available for use in subsequent steps of the ISD process.

3.9.1 Data sources. The most recent information available shall include, but not be limited to, the areas listed below to identify possible additions to, or modifications of, the objectives hierarchies:

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(a) Mission analysis data.

(b) Functions analysis data.

(c) Task listings/task listings update reports.

(d) Human factors engineering task analysis data.

(e) Fleet maintenance data.

(f) SMEs.

(g) Equipment ECPs.

(h) Phase maintenance requirements cards.

(i) MIMs.

(j) Manufacturers' maintenance instructions.

(k) Tactics changes.

(l) NATOPS changes.

3.9.2 Methodology. Once it has been determined that a change must be made to a portion of one of the objectives hierarchies, the effected portion(s) shall be modified or deleted, as necessary, in accordance with the procedures of 3.8. The revised objectives hierarchies shall be inserted with careful attention to insure appropriate reference numbering is used. If portions of objectives hierarchies are added to existing ones, the numbering shall be modified as necessary throughout to allow for the insertion.

3.9.3 Objectives hierarchies update report. Objectives hierarchies updates for each crew position (operator) or job category and level (maintenance) shall be documented as required in accordance with UDI-H-25715 and submitted to the contracting agency for review.

3.10 Training equipment survey. For weapon systems undergoing research, development, test, and evaluation (RDT&E), a survey shall be conducted to assess training equipment used in similar existing weapon systems and to assess the state-of-the-art in simulation for training. This survey shall begin early in the ISD process so that information collected is available for consideration during the development of functional specifications for training equipment. The survey shall be accomplished as described in the following paragraphs.

3.10.1 Similar weapon systems survey. Weapon systems that are similar to the new weapon system shall be identified. Similarity shall be determined on the basis of common characteristics, such as system type, mission(s), crew size, tactics, weapons, system/subsystems, and maintenance

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support. Similar systems identified, and the emerging system shall be described in terms of the above variables and any other relevant variables selected by the contractor. For each weapon system identified as similar, the model(s), manufacturer, date of manufacture, pending and projected ECPs, and degree of similarity to the emerging system on each variable shall be determined. This survey shall consider Navy, Marine, Air Force, and Army systems that are currently operational. All similar systems identified shall be reviewed for the purpose of selecting a smaller subset, reflecting the most similar and most recently introduced system.

3.10.2 Associated training equipment. For each system identified in 3.10.1 as most similar, all major training equipment that has been acquired to support training for its crew/maintenance personnel shall be listed. Major training equipment shall include range facilities, weapon system trainers (WSTs), tactics trainers, procedures trainers, maintenance trainers, and others as appropriate. For each piece of training equipment listed under each similar weapon system, the following information shall be developed:

(a) Device designation, model number, manufacturer, date of manufacture, and procuring agency.

(b) Missions, mission segments, and tasks (operator), or systems, subsystems, and tasks (maintenance) taught in the device.

(c) Learning objectives (training requirements) met in the device, along with evaluations of training effectiveness for each.

(d) Simulation features and their assessed training effectiveness.

(e) Instructional features and their assessed value to instructional processes.

(f) Performance measurement capability.

(g) Cost.

(h) ECPs and dates of ECPs.

(i) Facility requirements.

(j) Availability and reliability figures.

(k) Known problem areas, including constraints, design problems, and utilization problems.

(l) Indicated improvements or modifications.

(m) Utilization in terms of hours or number of events per week or month.

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(n) Problem or scenario control capabilities.

(o) Personnel and technical support requirements.

(p) The number of devices or media required for a stated student load.

These data shall be rigorously developed and placed in a format acceptable to any automatic data processors designated by the procuring activity. These data shall provide the basis of the commonality analysis in 3.10.3. Where possible, the results of objective trainer evaluations, certification studies, and quality assurance and reliability reports shall be obtained to form the above data. Subjective data from device users shall also be obtained wherever possible and included in the overall device assessment.

3.10.3 Operator commonality analysis. A commonality analysis shall be performed across all selected similar systems to determine the following information:

(a) The optimal number and mix of training equipment to compose a training device suite for a fleet weapon system training program.

(b) The optimal simulation features for each type of device.

(c) The optimal instructional features for each type of device.

(d) Common features which fail to support training (as well as common causes of failure).

Items (b) and (c) shall consider engineering change proposals, as well as original design. The development of the optimum suite of training equipment shall consider the availability of training and support facilities, quantity of operational equipment, and other assets identified in the PA. Sources for these data shall be the weapon system planning document and the PAR.

3.10.4 Subsystem equipment commonality analysis (maintenance). For maintenance applications, the equipment of all selected similar weapon systems shall be analyzed for instances of commonality with the weapon system under development. Wherever a particular subsystem, equipment unit, or equipment subunit is found to be common, information shall be collected regarding the similar system(s). Such information shall include, but not be limited to:

(a) Maintenance requirements.

(b) Maintenance experience.

(c) Maintenance procedures.

(d) Description of how hardware operates.

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(e) Maintenance training curricula.

(f) Maintenance training records.

(g) Maintenance problems.

The information shall be assembled to form a data bank, with provision for easy retrieval of specific items. Retrieval procedures shall be submitted for approval by the contracting agency. The equipment commonality analysis shall establish a reference source for use throughout the ISD process without the necessity of re-executing analyses already completed.

3.10.4.1 Methodology. The equipment commonality analysis for maintenance shall be conducted as explained below:

(a) Step 1: List equipment units per subsystem within weapon systems. A list of all equipment units and subunits per subsystem in the weapon system under consideration shall be developed.

(b) Step 2: Identify common units. A list of common items across all similar equipment shall be prepared.

(c) Step 3: Identify data sources. With respect to each common unit, all sources of information concerning areas listed in 3.10.4 shall be identified.

(d) Step 4: Obtain and catalog information from sources. The specified information from each source shall be collected and analyzed as required. Each acquisition shall be tagged and filed. An index to facilitate retrieval shall be established.

3.10.5 State-of-the-art assessment. The trainer industry, NAVTRAEQUIPCEN, and appropriate commands shall be surveyed to determine the existence of recently developed training equipment capabilities that exceed those surveyed in existing training programs. This survey shall emphasize features noted in 3.10.2 as being related to program or equipment deficiencies. The state-of-the-art survey shall result in the identification of new features to be added to those already found effective. A survey of new technology shall also be conducted in those areas of simulation or instructional features commonly found to fail to support training equipment requirements (3.10.3[d]). For each feature so identified, candidate replacement features shall be generated from the survey of new technology. Each new feature shall be identified in terms of nomenclature, source, risk, cost, indicated training capabilities, and expected training value.

3.10.6 Training equipment survey report. Procedures used to select and analyze similar weapon systems and associated training equipment and to conduct the state-of-the-art assessment shall be documented in accordance with UDI-H-25716 and submitted to the contracting agency for review.

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3.11 Media selection. Media selection shall be performed first for all hands-on objectives (those requiring real or simulated equipment). In addition to media selection for all hands-on objectives, this step shall result in the generation of functional specifications for new devices and recommendations for trainer modifications. Hands-on media selection shall be followed by academic media selection. The contractor shall select one of two approaches for selecting academic media: (1) media selection by objective, followed by syllabus development, and finalization of media by lesson or (2) syllabus development, followed by media selection by lesson. Alternate sequences of media selection and syllabus steps are shown in Figure 7. An explicit model for both hands-on and academic media selection shall be provided for approval by the contracting agency and, after approval, be used in carrying out both components of media selection. The media selection and syllabus development processes shall be documented after steps in both have been completed (Figure 7).

3.11.1 Media selection model report. A detailed description of the media selection model shall be documented in accordance with UDI-H-25717 and submitted to the contracting agency for review.

3.11.2 Hands-on media selection. Hands-on media selection shall be a three step process: (1) determination of all objectives requiring hands-on instruction/practice, (2) determination of the different types of trainers required to train the hands-on objectives, and (3) assignment of each objective to one or more of the trainer categories.

3.11.2.1 Identification of "hands-on" objectives. The objectives hierarchies shall be reviewed to identify all objectives which will require hands-on training/practice. This shall be accomplished by examining the behavioral statement and conditions for each objective. All objectives requiring real or simulated operational equipment shall be listed (i.e., objectives requiring visual, auditory, motion, environmental, and other cues which must be presented in some manner of live enactment).

3.11.2.2 Training equipment specifications. Once the objectives that require "hands-on" training have been identified, their common conditions shall be analyzed to establish trainer categories based on similar requirements. Trainer categories shall include, but not be limited to:

(a) Mission.

(b) Procedures.

(c) Familiarization.

(d) Troubleshooting.

Within the categories of trainers, consideration shall be given to crew/team vs. individual training, initial qualifications vs. refresher training, and fidelity requirements. The output of this step shall be military characteristics of each category of trainer.

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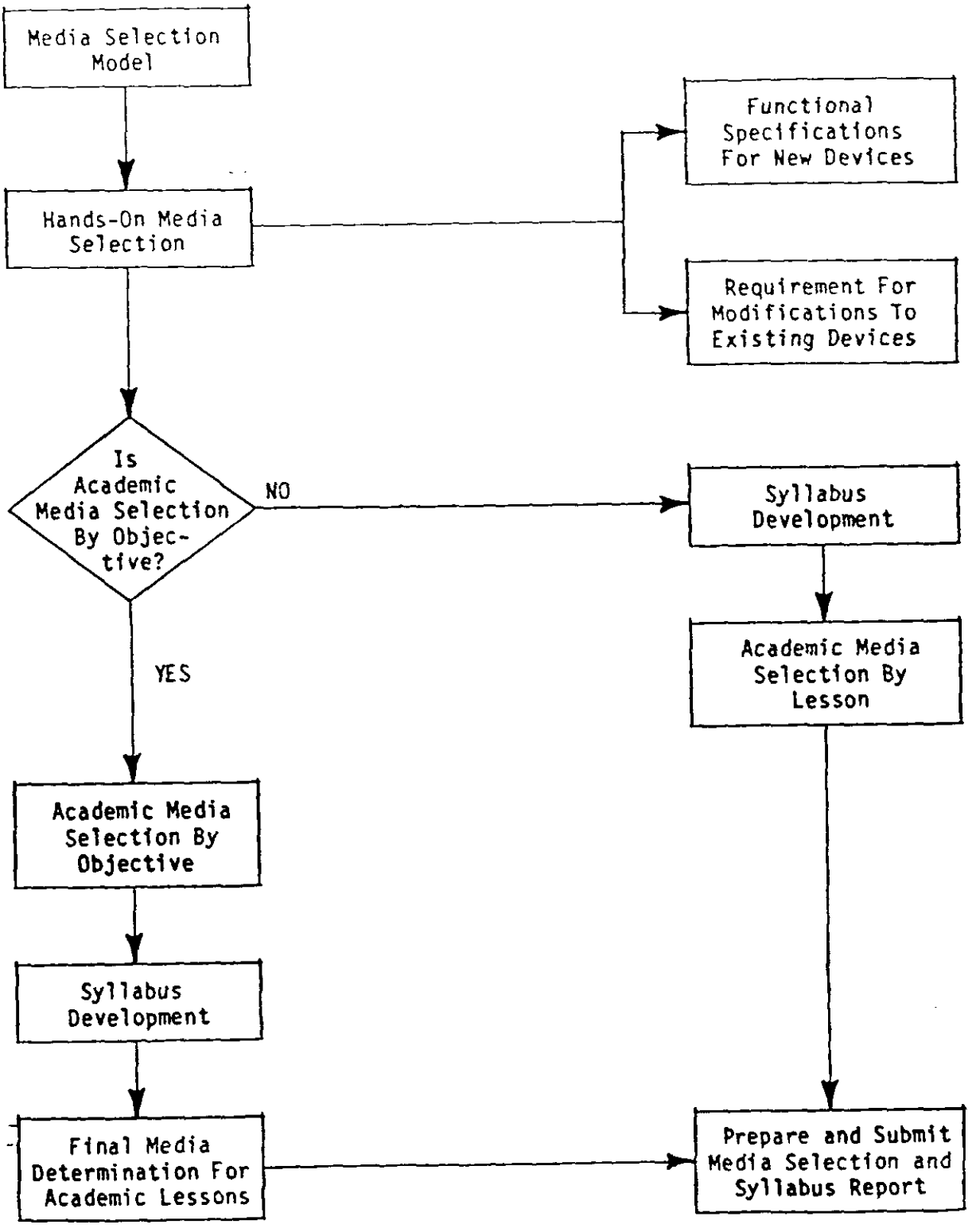


FIGURE 7. Alternative media selection and syllabus development sequences.

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3.11.2.3 Hands-on media selection by objective. Each objective requiring hands-on instruction/practice shall be compared with the features and capabilities of each category of training equipment in order to select the most suitable medium (or media). Selected media shall be adequate to handle all, or an acceptable number of, instructional requirements relative to each assigned objective. Specific considerations shall include ensuring necessary:

(a) Controls, dials, displays, etc., to support the accomplishment of required task behaviors.

(b) Simulated or actual operational/environmental or scenario conditions, including visual, proprioceptive, and auditory cues.

(c) Performance monitoring capabilities.

(d) Performance measurement capabilities.

(e) Operational or simulated scenario or problem control capabilities, including control over pacing.

(f) Tasks/objectives peculiar to maintenance, such as parts removal, adjustment, disassembly, and fault isolation and insertion.

One output of this step shall be primary and alternate (if appropriate) hands-on media for each objective identified as requiring hands-on instruction. The second output shall be a set of functional requirements for new devices/trainer modification requirements.

3.11.3 Trainer functional description report. Functional specifications for new devices for emerging and existing systems shall be documented in accordance with UDI-H-25718 and submitted to the contracting agency for review.

3.11.4 Trainer modification report. Recommended trainer modifications for existing systems shall be documented in accordance with UDI-H-25719 and submitted to the contracting agency for review.

3.11.5 Academic media selection. Academic media selection shall be either by objective or by lesson. Depending on the procedure used, the sequence of steps with respect to syllabus development shall be as shown in Figure 7.

3.11.5.1 Academic media selection by objective. After sets of acceptable hands-on media have been selected for objectives requiring hands-on instruction, a media analysis shall be conducted for each remaining objective. For each instructional objective requiring academic instruction, a media analysis shall be conducted in which primary and alternate media are selected for each objective. The decision points in the selection process shall be based on media-related learning variables, such as instructional

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display requirements, type of learning (e.g., recall, recognition, classification, etc.), student-program interaction level, and update requirements. The media available for selection shall reflect a full range of instructional capabilities and provide a full set of options within the constraints of program feasibility. The state-of-the-art shall be assessed to ensure that all relevant media have been considered. The output of this step shall be a list of acceptable media for each objective. Final media selection by lesson (i.e., determining if the same set of primary and alternate media was selected for each objective in a lesson) shall take place during syllabi development.

3.11.5.2 Academic media selection by lesson. If academic media selection is by lesson, course syllabi shall be developed prior to media selection. The subsequent media selection process shall consider objectives in each lesson as an aggregate. The decision points in the selection process shall be based on media-related learning variables, such as instructional display requirements, type of learning (e.g., recall, recognition, classification, etc.), student-program interaction level, and update requirements. The media available for selection shall reflect a full range of instructional capabilities and provide a full set of options within the constraints of program feasibility. The state-of-the-art shall be assessed to ensure that all relevant media have been considered. The output of this step shall be a list of acceptable media for each lesson.

3.11.6 SME training for media analysis. Since media selection is a contractor activity, no formal SME training shall be required. SME input shall be solicited as required during the selection process.

3.12 Operator syllabi. Depending on the method of media selection (i.e., by objective or by lesson), the syllabi development process shall be sequenced relative to media selection as shown in Figure 7. The output of syllabi development shall be a sequence of lessons for each training track in each student training course (STC). Lessons shall be composed of groupings of objectives from the objectives hierarchies. Primary and alternate media shall be specified for each lesson.

3.12.1 Determination of training tracks. Entry level skills of students from different sources (to include initial vs. refresher training) shall be determined from the PA and entry level behavior analysis. If there are sufficient differences among student populations, separate tracks shall be developed for each source of students.

3.12.2 Organization and sequencing. The general sequence to be used during course syllabus development is shown in Figure 8. The development of operator syllabi shall be conducted in accordance with the process outlined in the following paragraphs.

3.12.2.1 Step 1: Organize and sequence objectives which require operational equipment.

(a) In order to sequence segments requiring operational equipment, the following steps shall be followed:

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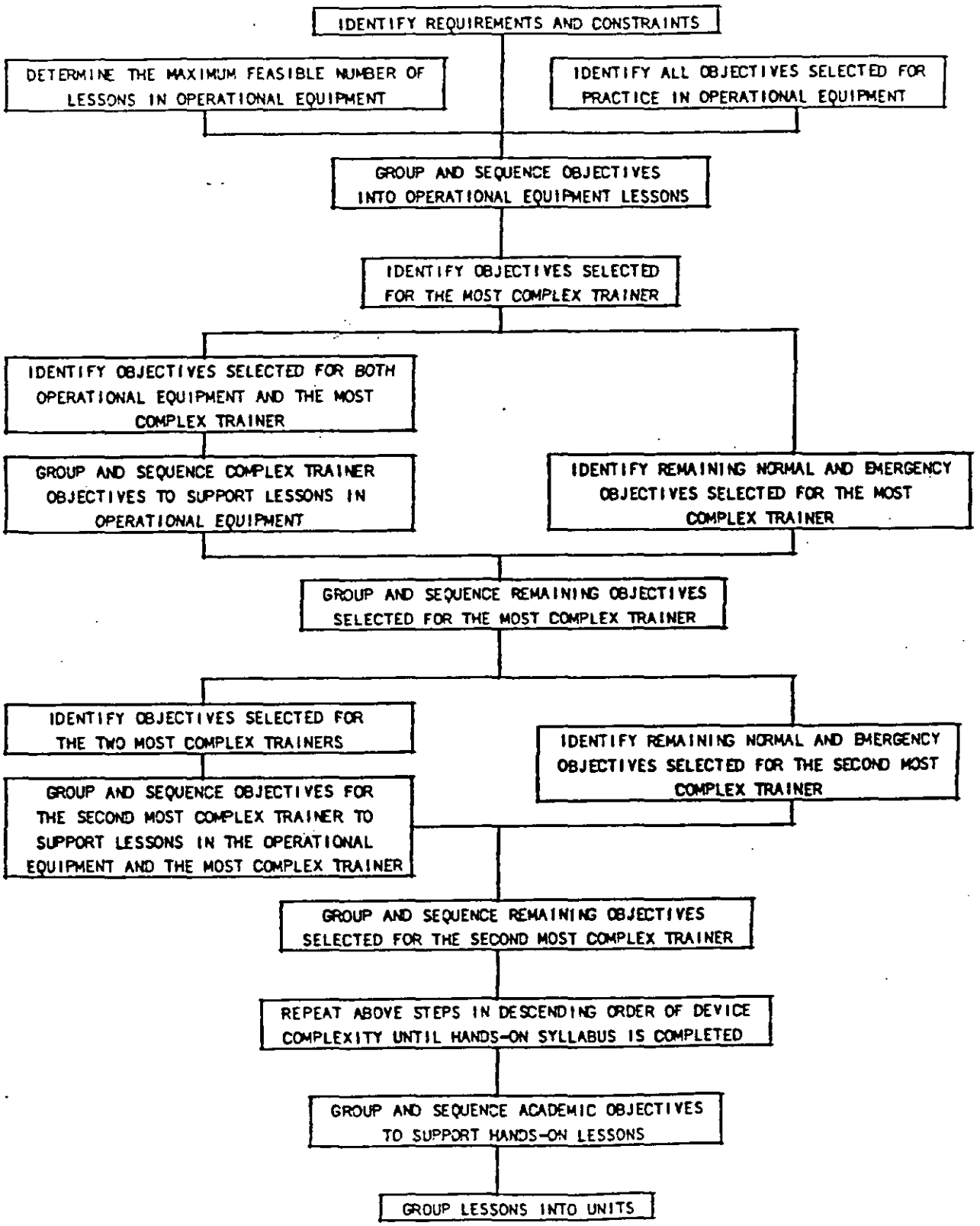


FIGURE 8. Syllabus development procedure.

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(1) Determine from PA and other available data (projected student throughput, number of pieces of operational equipment, projected average time per lesson on operational equipment) the maximum feasible number of syllabus lessons that can be conducted in the operational equipment.

(2) Identify any content requirements, ordering requirements, or other constraints which are imposed by the training activity or higher authority.

(3) Extract from the objectives hierarchies all objectives designated during media selection as requiring training in the operational equipment.

(4) Estimate the amount of practice in operational equipment required for each objective (consider training effectiveness and availability of simulators).

(5) Examine the objectives to identify common logical groupings.

(6) Sort the objectives into lessons in the operational equipment within imposed constraints. Develop only as many lessons as required to train the objectives, not to exceed (1) above.

(b) The following guidelines shall be considered when organizing lessons in the operational equipment:

(1) The first lesson should contain the absolute minimum number of objectives required to have a lesson at all.

(2) Subsequent lessons in the operational equipment should add new tasks so that the lessons proceed from easier to more difficult.

(3) Groupings should allow all crew members to practice their individual position objectives by complementing (or at least not interfering with) other crew members' performance objectives requirements.

(4) Every lesson should contain an appropriate number of objectives. This means that there should be sufficient objectives to warrant a lesson in the operational equipment but there should not be so many objectives that each cannot be properly evaluated.

(5) Every lesson should involve some new objectives which require student practice.

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(6) When a new objective is introduced, it should be practiced and evaluated in that lesson, in several of the immediately following flights, and then again at periodic intervals as needed to maintain competency.

(7) If possible, objectives should be grouped into lessons in the operational equipment such that the prerequisite classroom and trainer periods for each lesson form a set of instructional blocks which are reasonably unified in subject matter and of approximately equal size.

3.12.2.2 Step 2: Organize and sequence objectives to be practiced in trainers. The following procedures shall be followed:

(a) Lessons for trainers shall be developed in descending order of trainer complexity (i.e., the most complex trainer shall be considered first, followed by the next most complex trainer, etc.). The orientation shall be to make optimum use of each trainer in preparing students for lessons in more complex trainers and the operational equipment.

(b) When trainer capabilities permit, crew coordination exercises shall precede each lesson on the operational equipment. Objectives which are to be practiced in the operational equipment shall be practiced in the crew coordination exercises.

(c) When trainer capabilities permit, individual position trainer exercises shall be developed to support each crew trainer exercise.

(d) New trainer exercises shall be identified for those objectives which can be practiced and evaluated in the trainer but not in the operational equipment (i.e., certain emergency procedures). This type of objective shall be initially practiced and evaluated in separate trainer exercises, and then shall be interspersed throughout the rest of the trainer exercises to help maintain the student's proficiency.

(e) All trainer exercises shall be structured so that they are large enough to warrant a trainer (i.e., practice and evaluation require 1 or 2 hours of time), but small enough that all new behaviors can be adequately evaluated.

(f) In some circumstances, a series of exercises in a given trainer may be required prior to practice in the next most complex trainer.

(g) In order to organize and sequence lessons in the most complex trainer which will precede lessons in the operational equipment, the following steps shall be followed:

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(1) Identify objectives which, as a result of hands-on media selection, were assigned both to the most complex trainer and the operational equipment.

(2) Group these common objectives into lessons which prepare students for the following lesson on the operational equipment.

(3) Group and sequence the remaining objectives which were assigned to the most complex trainer but not to the operational equipment. These objectives may be assigned to lessons developed in (1) above or may constitute new lessons.

(h) In order to organize and sequence lessons in other trainers, which will provide prerequisite training for the most complex trainer and the operational equipment, the following steps shall be followed:

(1) Identify objectives which, as a result of hands-on media selection, were assigned to both the trainer under consideration and the next most complex trainer.

(2) Group these common objectives into lessons which prepare students for the following lesson on the next most complex trainer.

(3) Group and sequence the remaining objectives which were assigned to the trainer under consideration but not to the next most complex trainer. These objectives may be assigned to lessons developed in (1) above or may constitute new lessons.

(4) Execute (1), (2), and (3) above for each trainer.

3.12.2.3 Step 3: Organize and sequence academic objectives. In Steps 1 and 2 of the organization and sequencing process, a framework of operational equipment and trainer exercises was designed. All remaining objectives (i.e., academic objectives) shall be inserted into the framework and grouped into logical instructional blocks. These instructional blocks shall provide the prerequisite academic information for performance in the hands-on exercises. The following steps shall be followed:

(a) Using the objectives hierarchies, identify and key out all objectives that are prerequisite to the first trainer.

(b) Group these objectives into a series of lessons. A lesson:

(1) Is a set of approximately 3 to 15 objectives which can be adequately presented in one presentation medium.

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(2) Takes from 30 minutes to 2 hours to complete.

(3) Deals with a single subject matter topic.

(4) Encompasses enough material to result in a significant learning accomplishment.

(5) Demands a test to ensure that students have mastered the material and are ready to proceed further.

(c) Using the objectives hierarchies, identify and key out all objectives which, in addition to those already sequenced, are prerequisite to the next trainer or operational equipment exercise.

(d) Group these objectives into a lesson or series of lessons following the criteria for good lesson structure.

(e) Repeat steps (c) and (d) for each trainer and operational equipment exercise (in turn) until all objectives have been sequenced.

3.12.2.4 Step 4: Group lessons into units. The sequence of lessons shall be examined to identify logical units of instruction. The following criteria for identifying units shall be considered:

(a) Consists of a series of 1 to 15 lessons.

(b) Takes from 2 hours to 1 week to complete.

(c) Deals with a single subject matter area.

For purposes of scheduling, an operational equipment exercise is always the last lesson in a unit. In other words, any lesson following an operational equipment exercise is, by definition, the first lesson in a new unit unless, due to simulator constraints or practice requirements, multiple operational equipment events containing the same objectives are scheduled in sequence. A unit may contain no operational equipment lesson at all, but if it does, it shall be designated as the last lesson in the unit.

3.12.3 Selection of primary and alternate media for academic lessons. If academic media selection is by objective, each academic lesson shall be examined to determine primary and alternate media per lesson. Lesson content and context requirements versus media requirements of objectives shall be considered to determine if media requirements dictate lesson restructuring. The primary medium selected for each lesson shall have been identified during the determination of acceptable instructional media as an acceptable presentation medium for all objectives in that lesson. Lessons shall be restructured as required.

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3.12.4 Determination of media plans. Primary and alternate media selected for each lesson shall be examined to determine the preferred and alternate plans for syllabi presentation. Procedures shall be in accordance with 3.12.4.1 and 3.12.4.2.

3.12.4.1 Determine optimal trainer mix. The optimal number of each type of training device shall be determined for each STC. The following criteria shall be used to define an optimal trainer mix:

(a) Each trainer identified shall be scheduled at a significant usage rate such that it is scheduled for use 75 percent of available training time (i.e., normal training hours minus estimates of unscheduled maintenance).

(b) Subject to (a), all hands-on objectives shall be presented in the least expensive device possible that has the instructional capabilities required by the objectives.

3.12.4.2 Determine alternate academic media mixes. Primary and alternate media for academic lessons shall be examined to identify the primary and alternate academic media mixes. Consideration shall be given to, but are not limited to, the following:

(a) Update requirements of subject matter content and ease of update of candidate media.

(b) Range of costs.

(c) Pacing philosophy (i.e., individualized, self-paced, lockstep, or combinations).

(d) Student throughput.

(e) Constraints (e.g., instructor availability, support requirements, etc.).

The output of this analysis shall be preferred and alternate media mixes (media plans) for each STC.

3.12.5 Daily class schedule. A daily class schedule for the entire training system shall be developed. Any scheduling differences resulting from different media mixes shall be identified. A graphic layout shall be developed showing, at a lesson level, the daily class schedules for all concurrent courses of all STCs. This schedule shall be used to estimate course duration and resource requirements.

3.12.6 SME training for syllabi development. In order for SMEs to function adequately during the course organization and sequencing tasks, training shall be provided in the following areas:

(a) Familiarization with the course organization and sequencing process.

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(b) Structure of the objectives hierarchies.

(c) Acceptable limits for size of units and lessons.

(d) Identification of prerequisite relationships between objectives.

(e) Identification of logical testing points.

(f) Identification of logical groups of objectives.

(g) Use of accepted unit, lesson, and segment numbering system.

(h) Overview of media selection procedures and relationship between media selection and syllabus development.

(i) Media mix analysis procedures.

(j) Incorporation of external constraints.

(k) Application of this step to subsequent ISD steps.

(l) QC and review procedures.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.12.7 Media selection and syllabus report. Procedures used during media selection and syllabi development and syllabi for each STC, showing primary and alternate media, shall be documented in accordance with UDI-H-25720 and submitted to the contracting agency for review.

3.13 Maintenance syllabi. Depending on the method of media selection (i.e., by objective or by lesson), the syllabi development process shall be sequenced relative to media selection as shown in Figure 8. The output of syllabi development shall be a sequence of lessons for each training track in each student training course. Lessons shall be composed of groupings of objectives from the objectives hierarchies. Primary and alternate media shall be specified for each lesson.

3.13.1 Determination of training tracks. Entry level skills of students from different sources (to include initial vs. refresher training) shall be determined from the PA and entry level behavior analysis. If there are sufficient differences among student populations, separate tracks shall be developed for each source of students.

3.13.2 Organization and sequencing. An explicit syllabus development procedure shall be provided for approval by the contracting agency and, after approval, be used in carrying out this task. The procedure shall include, as a minimum, the elements in 3.12.2 tailored to project-specific requirements.

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3.13.3 Other maintenance syllabi steps. Other procedures to be used in the development of maintenance syllabi shall be as specified in 3.12.3 through 3.12.7.

3.14 Syllabi update. The syllabi shall be updated to ensure that the most current data possible is available for use in subsequent ISD steps. In addition, media updates shall be made as required. Media updates/changes which impact media plans shall require a media plan update. Syllabi and media updates which will affect resource requirements for subsequent ISD phases shall be used to develop/update the TSRA.

3.14.1 Data sources. The most recent information available in the areas listed below shall be used to identify possible modifications to the syllabi and media:

(a) Task listings update reports.

(b) Objectives hierarchies update reports.

(c) QC data.

3.14.2 Methodology. Syllabus and media updates shall be carried out as described in the following paragraphs.

3.14.2.1 Syllabus changes. Once it has been determined that a change must be made to a syllabus, the effected portion(s) shall be modified as necessary, in accordance with procedures of 3.12 and 3.13.

3.14.2.2 Media updates. For those syllabus changes requiring media selection/updates, the same media selection procedures used during initial media selection shall be implemented.

3.14.2.3 Media plan update. After syllabus and media updates have been made, the media plan shall be examined to determine if there is a need for update. If a need exists, the media plan shall be updated.

3.14.3 Media selection and syllabus update report. Media and syllabus updates for each training syllabus shall be documented as required in accordance with UDI-H-25720 and submitted to the contracting agency for review.

3.15 TSRA. The TSRA shall be conducted to determine total personnel, services, materials, and facilities required to design, develop, implement, and maintain the proposed training syllabi for the weapon system. A complete analysis shall be conducted for each media plan.

3.15.1 Methodology. The following procedure shall be used to perform the TSRA.

3.15.1.1 Determine contractor and government roles. The TDSP and other relevant documents/policies shall be reviewed to determine contractor and government roles in each phase (e.g., contractor or government SMEs,

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contractor or government printing, etc.). For those roles and responsibilities which are not clearly defined, relevant organizations/personnel shall be consulted for definitization of these roles and responsibilities. Separate contractor and government resource requirements shall be estimated for each media plan in each phase.

3.15.1.1.1 Contractor resources. All contractor resources shall be estimated in terms of quantities per resource (e.g., man-hours, number of pieces of equipment, amount of material, etc.) and cost. Labor rates shall reflect average current market rates, to include base rates plus overhead and profit. Other costs shall reflect average commercial prices, plus any overhead.

3.15.1.1.2 Government resources. All government resources shall be estimated in terms of quantities and cost. Within costs, distinction shall be made between costs for which funds must be allocated (e.g., purchase of equipment, TAD assignment of personnel, reproduction of materials) and those for which funds need not be allocated (e.g., SMEs located on-site, available furniture, utilities, etc.).

3.15.1.2 Determine design resource requirements for each STC. The following categories of resource requirements shall be determined for the design activities of each candidate media plan.

(a) Personnel: The types of personnel, the total man-hours needed for each type, and costs shall be specified.

(b) Facilities and equipment: The office space, furnishings, and office equipment requirements shall be specified.

3.15.1.3 Determine development resource requirements for each STC. The following categories of resource requirements shall be determined for the development activities of each candidate plan.

(a) Personnel: The types of personnel, the total man-hours needed for each type, and costs shall be specified.

(b) Facilities: The office space and furnishings requirements shall be specified.

(c) Equipment: The equipment requirements (e.g., cameras, photocopiers, printing presses, slide trays, etc.) shall be specified.

(d) Materials/services: The raw materials (e.g., paper, photographic film, videotape footage, etc.) and services (e.g., film processing, printing, etc.) shall be specified.

3.15.1.4 Determine implementation resource requirements. If so indicated in the TDSP, distinctions shall be made between requirements during the first year of implementation, during which contractor support will most

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likely be available to assist in start-up and evaluation activities, and the remaining years in the life cycle of the training system when all training system operations and evaluation will be carried out by the government. The following categories of resource requirements shall be determined for the implementation activities of each candidate media plan. Care shall be exercised to insure resource requirements specified are adequate for training during periods of maximum student load.

(a) Personnel: Personnel requirements shall be identified. At a minimum, personnel requirements for the following activities shall be determined:

(1) Instructing.

(2) Scheduling.

(3) Counseling.

(4) Proctor/monitoring.

(5) Lecturing.

(6) Equipment/materials maintenance/distribution.

(7) Other administration.

(8) Evaluation.

(b) Equipment: Total equipment requirements (e.g., trainers, aircraft, carrels, etc.) shall be specified.

(c) Facilities: Facilities requirements in the following categories shall be identified:

(1) Learning center for individualized instruction (i.e., size, contents) shall be specified.

(2) Classrooms for lectures.

(3) Equipment/media/materials storage.

(4) Trainer and operational equipment related facilities.

(5) Administrative and office facilities.

3.15.1.5 Determine the yearly resources required for maintenance, revision, and update. Distinctions shall be made between requirements during the first year after implementation when revision requirements are greatest and during the remaining years in the life cycle of the training system. If so indicated in the TDSP, separate requirements shall be estimated for years

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in which contractor support will be available and years in which all maintenance, revision, and update activities will be carried out by the government. The yearly resources required for maintenance, revision, and update shall be specified:

(a) Personnel: The personnel requirements in excess of those required for implementation (3.15.1.4[a]) shall be specified.

(b) Facilities: The office space and furnishings in excess of those required for implementation (3.15.1.4[c]) shall be specified.

(c) Equipment: The equipment requirements (e.g., cameras, photocopiers, printing presses, slide trays, etc.) shall be specified.

(d) Materials/services: The raw materials (e.g., paper, photographic film, videotape footage, etc.) and services (e.g., film processing, printing, etc.) shall be specified.

3.15.2 SME training for TSRA. Since the TSRA is a contractor activity, no formal SME training shall be required. SME input shall be solicited as required during the analysis process.

3.15.3 Training support requirements report. Procedures used to conduct the TSRA and a detailed analysis of all resource requirements necessary to design, develop, implement, and maintain the training program shall be documented in accordance with UDI-H-25721 and submitted to the contracting agency for review.

3.16 TSRA update. In the event that all media plans considered in the TSRA are judged to be unfeasible or the selected media plan is updated during subsequent ISD phases, a revision of the TSRA shall be required.

3.16.1 Methodology. Since the nature and scope of required revisions cannot be predicted, it is not possible to provide details that specify how the revision process should be conducted. The general procedure, however, shall be the same as was used during initial TSRA preparation.

3.16.2 TSRA update report. TSRA updates shall be documented in accordance with UDI-H-25721 and submitted to the contracting agency for review.

3.17 STC lesson specifications. The output of the lesson specification development step of ISD shall be detailed outlines of all lessons specified during the development of the STC syllabi. The procedures used shall be a function of the instructional medium selected for each lesson. Specifications for academic lessons shall be developed as described in 3.17.1. Specifications for hands-on lessons (i.e., trainers and the actual equipment) shall be developed as described in 3.17.2. In order to ensure STC continuity and to cross-check syllabus sequencing and content, the order in which

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lesson specifications are developed shall correspond to the order of the lessons in the syllabus.

3.17.1 Academic lesson specification development. A lesson specification shall be developed for each academic lesson. Each lesson specification shall contain both general lesson information and a segment specification for each objective in the lesson. General lesson information shall include the following:

(a) Lesson reference number (course, unit, and lesson number within the unit).

(b) Lesson title.

(c) Lesson map (i.e., a hierarchical diagram showing prerequisite relationships among lesson objectives).

(d) Lesson objectives with reference numbers.

(e) Lesson introduction.

(f) Prerequisite relationships of the lesson (unit lesson map).

(g) Resources required to conduct the lesson (i.e., audiovisual components, worksheets, checklists, etc.).

(h) Student preparation requirements.

(i) Strategy for presenting and testing subject matter contents (e.g., whole - part - whole, discovery method, etc.).

3.17.1.1 Segment specification development. A segment is defined as the instruction required to teach one academic objective. Each segment specification shall contain the critical content and details of the instructional strategy necessary to teach the objective. Components of each segment specification shall include the following information relative to teaching one objective:

(a) Segment reference number (i.e., course, unit, lesson, and sequential segment number within the lesson).

(b) Segment title.

(c) Segment objective.

(d) Fact, category, rule, procedure, or principle statement.

(e) Instructional strategy.

(f) Help (description or actual help).

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(g) Instance specification.

(h) Example items.

(i) Practice and test items.

(j) Graphics specifications.

The segment specification development process for STC lessons shall be conducted as described in the following paragraphs.

3.17.1.1.1 Develop fact, category, rule, procedure, or principle statement. The fact, category, rule, procedure, or principle statement shall be developed first. This section contains the subject matter content of the objective. The particular statement that is developed for a single objective depends on the type of objective. For purposes of conciseness, the fact, category, rule, procedure, or principle statement shall be incorporated under the single term "generality." The contents of each of these types of generalities shall be as follows:

(a) A fact statement is information which must be remembered by the student in order to meet the requirements of the objective. Facts are simple associations between names, objects, symbols, locations, etc. They require recall or recognition, but not use of the information as in the other four types of generalities. This type of statement shall be used if the objective requires the student to: "state the dimensions of . . .," "recall the capacity of . . .," etc.

(b) A category statement is a list of the critical characteristics of a concept. It requires the student to remember the characteristics or to classify objects, events, or ideas according to the characteristics. This type of statement shall be used if the objective requires the student to: "identify examples of . . .," "determine whether it is an example of . . .," etc.

(c) A procedure statement is a sequence of steps remembered or used in a single situation or on a single piece of equipment. The conditions initiating the procedure at different times are the same or similar and the results are the same. This type of statement shall be used if the objective requires the student to: "set up the . . .," "perform a _____ maneuver," etc.

(d) A rule statement is a sequence of steps remembered or used which apply across situations or across equipments. It identifies the relationship among concepts and specifies the proper actions. Learning must involve both recognizing the conditions governed by the rule and applying the rule. The inputs and outputs from the rule may vary across a wide range. This

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type of statement shall be used if the objective requires the student to: "calculate," "determine," "solve for a value," etc.

(e) A principle statement involves remembering, interpreting, or predicting why or how things happen, or cause-effect relationships. This type of statement shall be used if the objective requires the student to: "explain how," "explain why," etc.

In order to develop the generality, the following general steps shall be followed:

(a) Step 1. The objective shall be reviewed to identify precisely what is being required of the student and to specify the type of generality (i.e., fact, category, rule, procedure, or principle).

(b) Step 2. Information related to the objective shall be collected from sources such as:

(1) Technical manuals.

(2) Existing training materials.

(3) Interviews with other SMEs.

(c) Step 3. The collected information shall be analyzed to determine:

(1) What is to be memorized for fact statements.

(2) The critical attributes of the concept for definition statements.

(3) The formula or steps to be followed for the rule statements.

(4) The steps to be followed for the procedure statements.

(d) Step 4. A rough draft of the generality shall be prepared.

(e) Step 5. The draft shall be reviewed to make sure that:

(1) All information necessary to accomplish the objective is present.

(2) All information that is not essential to accomplish the objective is excluded.

(3) As much as possible, terminology used will be familiar to the student. New terms will be concisely defined.

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(4) The wording and grammar are clear, concise, and easy to follow.

(f) Step 6. The objective and generality shall be reviewed with another author to get his/her comments with respect to whether or not they are satisfactory.

(g) Step 7. The generality shall be revised as necessary to meet the criteria outlined in step 5 above.

(h) Step 8. A smooth version of the generality shall be prepared.

(i) Step 9. As generalities are developed, they shall be reviewed by other SMEs. They shall also be compared with generalities developed earlier in the same course (and in other courses) to make sure that all generalities have internal consistency.

3.17.1.1.2 Prepare remaining portions of STCs segment specification. Once the generality has been completed, development of the remaining portions of the segment specification shall begin. These sections shall contain the strategy and procedures for teaching the generality. The contents of each of the remaining sections of the segment specification shall be as follows:

(a) Instructional strategy. This section of the segment specification shall outline the instructional strategies to be used in presenting, practicing, and testing the generality. The strategies shall be sufficiently clear and concise to enable authors of the instructional materials to incorporate the strategies without confusion or error.

(b) Help. This section of the segment specification shall outline the type of material to be presented to help the student remember and use the generality. Methods of providing this help include:

(1) Giving a mnemonic (memory trick).

(2) Giving a general example of how the statement can be used.

(3) Explaining why the statement is important.

(4) Explaining how it came about, how it fits in the course, or how it relates to something the student already knows.

(5) Explaining some of the terms in the statement.

(6) Representing the statement with pictures, symbols, flowcharts, tables, etc.

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Types of help for different generalities are as follows:

(1)	Helps for categories:	Highlight the critical characteristics of an example. Explain why or why not something is classified as a member of a concept. Show the use of a checklist or heuristic to help classify. Simplify early examples (e.g., use line drawings instead of complicated photographs).
(2)	Helps for procedures or rules:	Explain why each step is done. Explain why each step is important. Give additional information about how to perform the task. Give additional information about how to know if you've done it wrong. Give flowcharts, tables, etc.
(3)	Helps for principles:	Highlight important features. Simplify the relevant information from the case study in which it is embedded. Use logical representations of the if-then relationships. Give additional information about how the principle applies, or why it doesn't. Give hints as to how to analyze problems.

(c) Instance specifications. This section shall contain information to be used in the development of example, practice, and test items. It shall consist of a common error analysis (CEA), descriptions of types of items to be used, and the number of items of each type.

(1) CEA. The CEA shall outline the common logical errors the student is likely to make in working practice and test items. It may also contain factual information which may be confused with information in the generality. The CEA will assist in the development of instances which teach students to discriminate among similar, potentially confusing instances (e.g., two different emergencies may have all but one initiation cue in common and, therefore, be easily confused).

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(2) Instance type. Each type of instance shall be described. Instance types shall include the full range of possible situations feasible in the training and operational environments. Both examples and non-examples shall be included (i.e., situations which are representative of the information/cues/procedures contained in the generality and those which are similar and could be confused with the generality content). Different display formats shall be considered (e.g., pictorial and verbal, verbal, etc.). Instance types shall be consistent with the medium used for the lesson. If practice/testing in another medium is required (e.g., simulator practice after academic instruction), the instance type shall be described and referenced to the appropriate lesson.

(3) Instance production. The number of example, practice, and test items for each type of instance shall be provided. The number and type of example, practice, and test instances shall be sufficient to ensure proper understanding of the generality, provide a full range and amount of practice, and allow accurate evaluation of student understanding/application of the generality.

(d) Example items. This section shall contain the example items to support instruction of the generality. Most fact statements will not require examples. Category, procedure, rule, and principle statements will require examples. Care shall be taken to ensure that the examples are representative of the generality and the job, and that they cover the entire range of possible examples. The set of example items shall contain the number and types of items described in the instance specifications.

(e) Practice items. This section shall contain the practice items to support instruction of the generality. Feedback for all practice items shall be provided. Care shall be taken to ensure that the practice items are representative of the generality and that they cover the entire range of possible applications of the generality. The set of practice items shall contain the number and types of items described in the instance specifications.

(f) Test items. This section shall contain the test items for the generality. Sufficient items shall be developed to enable two versions of the lesson test. The set of test items shall contain the number and types of items described in the instance specifications. The number and format of test items shall ensure that:

(1) They are clear and unambiguous.

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(2) They do not give away answers among items.

(3) They are well constructed.

(4) Their type(s) is keyed to the characteristics of the objective and generality (e.g., recall, recognition, calculation with fill-in response).

(5) Students cannot successfully pass them by guessing.

(6) They measure the behavior specified in the objective.

(g) Graphics specifications. This section shall contain sketches and descriptions/references to illustrations which will be used to provide the graphical portions of instruction. Depictions and verbal descriptions shall be of sufficient clarity to ensure unambiguous communication of content with authors and IDs. References shall be complete and accurate.

3.17.2 Hands-on lesson specification development. A lesson specification shall be developed for each lesson to be conducted on a trainer or the actual equipment. Each specification shall contain the essential elements required for student practice and instructor evaluation of the objectives in the lesson.

3.17.2.1 Methodology. Hands-on lesson specifications shall be developed and formatted using procedures and conventions provided to the contracting agency for approval. At a minimum, hands-on lesson specifications shall consist of the following components:

(a) Lesson reference number (i.e., course, unit, and lesson number within the unit).

(b) Lesson title.

(c) Introduction (explaining the goals of the lesson and relating the lesson to previous and future lessons).

(d) Lesson objectives.

(e) Resources required to conduct the lesson (i.e., student guide, evaluation checklist, etc.).

For each objective in a hands-on lesson, the following shall be provided:

(a) Reference to the academic lesson(s) in which instruction on the objective is presented. This will provide a cross-check to ensure that relevant, enabling information is contained in the syllabus and is sequenced prior to first performance of the objective. It will also facilitate student review in preparation for hands-on lessons.

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(b) Conditions under which the objectives will be performed (from the task listing and objectives hierarchies). Care shall be taken to ensure that conditions are complete and accurate.

(c) Standards of performance. Precise standards based on the measurement variables specified in the objectives hierarchies shall be provided. Where appropriate, quantitative values of the measurement variables shall be specified. In developing standards for objectives practiced in more than one lesson, levels of student proficiency, based on amount of practice, shall be considered (i.e., standards would be expected to be less rigorous during early practice than during later practice). For all objectives, terminal standards shall reflect realistic proficiency levels required upon assignment to an operational squadron.

3.17.3 Student evaluation procedures. After lesson specifications have been developed, academic standards, test specifications, and hands-on performance standards shall be reviewed. This information, along with general evaluation procedures and philosophies from the fields of educational psychology, instructional technology, etc., shall be used to specify general evaluation procedures to be used for academic and hands-on lessons in each STC. Elements of evaluation to be considered include, but are not limited to:

(a) Number of different types of test items to be used within and across lessons.

(b) Scoring criteria.

(c) Remediation and recycling philosophies.

(d) Use of pretests.

(e) Development and use of rating scales.

(f) Test points within each STC.

(g) Training activity requirements (e.g., tests, check-rides, etc.).

3.17.4 SME training for lesson specification development. In order for SMEs to function adequately during lesson specification development, training shall be provided for both academic and hands-on lessons.

3.17.4.1 Academic lesson specification development. Training for the development of academic lesson specifications shall cover the following areas:

(a) Development of each of the five types of generalities.

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(b) Development of helps.

(c) Development of example items.

(d) Development of practice items.

(e) Development of test items.

(f) Development of graphics specifications.

(g) Development of standards.

(h) Development of lesson introductions.

(i) Development of lesson maps.

(j) Determination of evaluation procedures.

(k) Use of proper lesson and segment specification formats.

(l) Use of instructional strategies.

(m) QC and review procedures.

3.17.4.2 Hands-on lesson specification development. Training for the development of hands-on lesson specifications shall cover all areas of the procedure proposed by the contractor and, at a minimum, shall include the following:

(a) Procedures for identifying and formatting references.

(b) Development/refinement of conditions.

(c) Development of standards.

(d) Development of lesson introductions.

(e) Determination of evaluation procedures.

(f) Use of proper lesson specification formats.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.17.5 Lesson specification report. Procedures used during lesson specification development, lesson specification formats, and specifications for each lesson in each STC shall be documented in accordance with UDI-H-25722 and submitted to the contracting agency for review.

3.18 Lesson specification update. The lesson specification shall be updated as required to insure that the most current data possible is available for use during subsequent ISD steps.

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3.18.1 Data sources. The most recent documentation available shall include, but not be limited to, the areas listed below to identify needed revisions to the lesson specifications:

(a) Engineering design data.

(b) System test data.

(c) Logistics support analysis data.

(d) Objectives hierarchies update reports.

(e) STCs' syllabi update reports.

(f) STC lesson materials formative tryout data.

(g) Technical publications.

(h) Fleet maintenance data.

3.18.2 Methodology. Once it is determined that a change must be made to a lesson specification(s), the affected specification(s) shall be modified as necessary. Cross-checks shall be made to ensure that updates are made to all lessons affected by a change (e.g., the addition of a new objective to a hands-on lesson may require update of an academic lesson, a change in method of performance of a hands-on objective may require updates of both academic and hands-on lessons, etc.).

3.18.3 Lesson specification update report. Lesson specification updates for each STC shall be documented as required in accordance with UDI-H-25722 and submitted to the contracting agency for review.

3.19 Lesson formatting. The output of the lesson formatting step of ISD shall be a set of guides which describe exactly how the material in each lesson specification will be expanded, developed, and formatted to produce the actual lesson of instruction. It shall explain how the individual segments shall be tied together to form academic lessons and how information on individual objectives in hands-on lessons shall be used to structure trainer and actual equipment lessons. A separate lesson format guide shall be developed for each medium.

3.19.1 Methodology. For each instructional medium, procedures for using the information contained in each component of a lesson specification to author lesson materials shall be provided. Procedures presented shall clearly relate the contents of the lesson specifications to the contents of authored materials and tests. To support the presentation of procedures, a minimum of one example per medium shall be provided. Examples shall be taken from the lesson specifications for the STC being developed and shall contain both the lesson specification and the authored materials and tests, with narrative explaining the transition from the former to the latter.

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3.19.2 Lesson format guides. The lesson format guides shall be documented in accordance with UDI-H-25723 and submitted to the contracting agency for review.

3.20 Lesson materials and tests. The output of this ISD step shall be lesson materials and tests for each lesson in each of the STCs. This phase of development shall be made up of the following five interrelated tasks: authoring, prototype production, small-scale tryout, revision, and final production. The general flow of tasks during lesson authoring and tryout is illustrated in Figure 9.

3.20.1 Methodology. The development of the STCs lesson materials and tests shall be conducted as described in the following paragraphs.

3.20.1.1 Step 1: Author paper and pencil version of lesson materials and tests. An initial version of each lesson shall be authored using the input of the lesson specifications and lesson format guides. The content of the lesson shall be as prescribed in the relevant lesson specification. The format of the materials shall be as prescribed in the lesson format guide for that presentation medium. The actual "pieces" of material that are produced during this step for lessons in representative media shall be as outlined in Figure 10. Several interactive content reviews by other SMEs and individual tryouts by "naive" students shall be included during this step to ensure that the paper and pencil lessons are as technically accurate and comprehensive as possible.

3.20.1.2 Step 2: Prototype production of lesson materials and tests. Once the paper and pencil version of a lesson has been authored, a prototype of the lesson shall be produced for use during small-scale tryouts. The exact procedure used to produce a lesson is highly variable, depending on the organization of the production shop and the presentation medium. Whatever the specific production model used, it shall incorporate sufficient content review, graphics review, editing review, and proofreading to ensure that the prototypes are as free as possible from content and format problems. The actual "pieces" of instruction produced during this step for lessons in representative media shall be as outlined in Figure 10.

3.20.1.3 Step 3: Small-scale tryout of prototype materials. Once the prototype version of the lesson materials has been produced, it shall be evaluated in a small-scale tryout. The procedures followed during the small-scale tryout of the prototype materials are described in more detail in 3.21.

3.20.1.4 Step 4: Revision of prototype lesson materials and tests. Once the small-scale tryout data has been analyzed, necessary revisions shall be made to the prototype lesson materials as dictated by the data. If a major revision is called for, the surrounding related lessons, the lesson organization, and the objectives hierarchies shall be examined to determine the effect that the revisions will have on them. Modifications may have to be made in one or more of these areas to ensure consistency is maintained throughout the course. If major revisions are made to the lesson, it shall be submitted to a second prototype production and small-scale tryout cycle.

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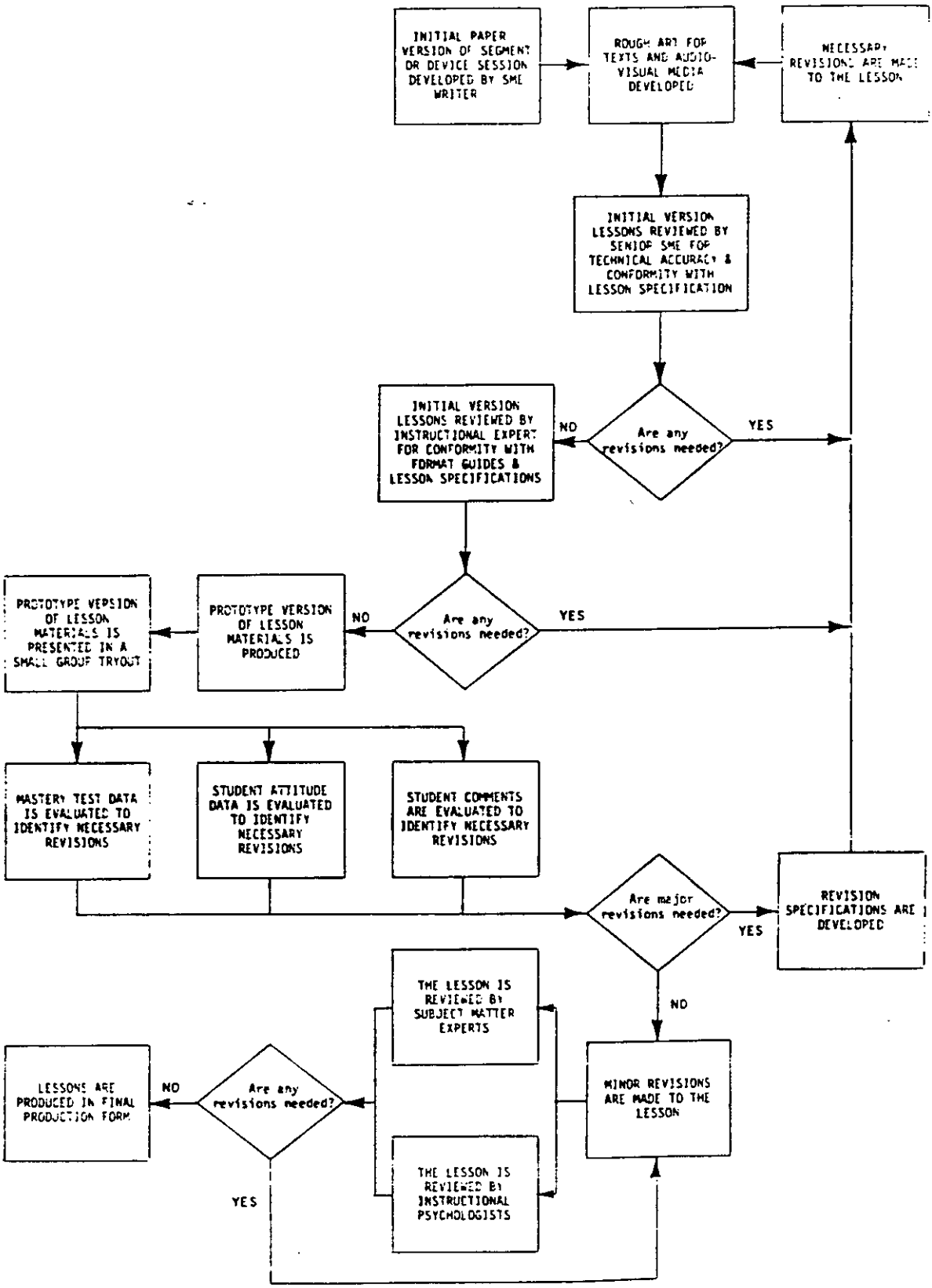


FIGURE 9. Lesson authoring and formative evaluation process.

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MEDIUM	COMPONENT	FORM OF PRODUCTS AFTER AUTHORING	FORM OF PRODUCTS AFTER PROTOTYPE PRODUCTION	FINAL FORM OF PRODUCTS
MEDIATED INTERACTIVE LECTURE	1) LECTURE OUTLINE	TYPED	TYPED	TYPED
	2) STUDENT WORKSHEET (INCLUDING TESTS)	TYPED WITH SKETCHES OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED OR BETTER REPRODUCIBLE MASTERS
	3) LECTURE AIDS	SKETCHES AND/OR WRITTEN DESCRIPTIONS OF SLIDES, OVERHEADS, MODELS, ETC.	SKETCHES OF SLIDES, OVERHEADS, MODELS, ETC.	FINISHED REPRODUCIBLE MASTERS OF SLIDES, OVERHEADS, MODELS, ETC.
WORKBOOK	1) WORKBOOK (INCLUDING TESTS)	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED OR BETTER REPRODUCIBLE MASTERS
SLIDE-TAPE PRESENTATION	1) SLIDES	SKETCHES AND/OR WRITTEN DESCRIPTIONS AS PART OF A STORYBOARD	SKETCHES OF SLIDES	FINISHED REPRODUCIBLE MASTERS
	2) AUDIO-TAPE	TYPED - AS PART OF A STORYBOARD	TYPED SCRIPT	RECORDED PULSED REPRODUCIBLE MASTER CASSETTES
	3) STUDENT WORKSHEET (INCLUDING TESTS)	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED (OR BETTER) FORMATTED REPRODUCIBLE MASTERS

FIGURE 10. Sample list of products during each step of lesson authoring.

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MEDIUM	COMPONENT	FORM OF PRODUCTS AFTER AUTHORING	FORM OF PRODUCTS AFTER PROTOTYPE PRODUCTION	FINAL FORM OF PRODUCTS
RANDOM-ACCESS SLIDE PRESENTATION	1) SLIDES	SKETCHES AND/OR WRITTEN DESCRIPTIONS OF SLIDES	SKETCHES OF SLIDES	FINISHED REPRODUCIBLE MASTERS
	2) STUDENT WORKSHEET (INCLUDING TESTS)	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED (OR BETTER) FORMATTED REPRODUCIBLE MASTERS
VIDEOTAPE PRESENTATION	1) VIDEOTAPE	AUDIO TYPED AND VIDEO SKETCHES AND/OR DESCRIPTIONS IN A STORYBOARD FORM	TYPED STORYBOARD WITH SKETCHES	FINISHED REPRODUCIBLE MASTER CASSETTE VIDEOTAPE
	2) STUDENT WORKSHEET (INCLUDING TESTS)	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED (OR BETTER) FORMATTED REPRODUCIBLE MASTERS
HANDS-ON EXERCISES	1) TRAINER SOFTWARE (IF NECESSARY)	TYPED COPY OF DESCRIPTION OF WHAT THE SOFTWARE MUST DO TO ALLOW THE STUDENT TO PERFORM THE OBJECTIVES	FIRST VERSION OF ACTUAL SOFTWARE IMPLEMENTED ON THE DEVICE	FINAL DEBUGGED VERSION OF ACTUAL SOFTWARE
	2) STUDENT WORKSHEET	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED (OR BETTER) FORMATTED REPRODUCIBLE MASTERS
	3) STUDENT EVALUATION SHEET	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED (OR BETTER) FORMATTED REPRODUCIBLE MASTERS
	4) STUDENT FEEDBACK SHEET	TYPED WITH SKETCHES AND/OR WRITTEN DESCRIPTIONS OF FIGURES	TYPED WITH FIGURES DRAWN	TYPED (OR BETTER) FORMATTED REPRODUCIBLE MASTERS

FIGURE 10. Sample list of products during each step of lesson authoring. - Continued

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3.20.1.5 Step 5: Final production of lesson materials and tests. After a lesson has been through the tryout and revision cycle, it shall be sent to final production. As with prototype production, the exact final production procedure will vary with the production facility and the medium of the lesson being produced. A QC system shall be implemented to insure adequate content review, graphics review, editing, and proofreading review. A list of actual final products for lessons in representative media shall be as outlined in Figure 10.

3.20.2 SME training for lesson authoring. In order for SMEs to function adequately during lesson authoring, training shall be provided in the following areas:

(a) Familiarization with the lesson authoring process.

(b) Interpreting lesson specifications.

(c) Interpreting lesson format guides.

(d) Refinement and integration of helps.

(e) Refinement and integration of example sets.

(f) Refinement and integration of practice items.

(g) Refinement, format, and organization of test sets.

(h) Scripting techniques for audio-tape, slide-tape, videotape, workbook presentations:

(1) Proper use of visuals.

(2) Proper audiovisual coordination.

(3) Proper use of pauses and pace.

(i) Development of hands-on lesson materials.

(1) Student worksheets.

(2) Student evaluation sheets.

(3) Instructor guides.

(4) Other.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.20.3 SME training for formative evaluation. In order for SMEs to function adequately during formative evaluation of lesson materials and tests, training shall be provided as per 3.21.7.1.

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3.20.4 Student training materials. Final lesson materials and tests shall be delivered in accordance with UDI-H-25724.

3.20.5 Training materials development report. Procedures used during the development of lesson materials, an accounting of the quantities of materials produced, and manpower expended during the production phase shall be documented in accordance with UDI-H-25725 and submitted to the contracting agency for review.

3.21 QC plan. A plan for continuously assessing the effectiveness, efficiency, and palatability of the instructional materials and the instructional management system shall be developed for approval by the contracting agency and, after approval, implemented. The QC plan shall provide for the assessment of the instructional management system and the instructional materials from the earliest materials development stage to implementation of training activities and throughout the life cycle of the weapon system. It shall provide evaluation procedures which determine instruction requiring modification and revision procedures which bring about the required modifications. The QC plan shall address three major areas: formative evaluation, internal QC, and external QC.

3.21.1 Formative evaluation planning methodology. Formative evaluation is the process of review and tryout of instructional methods and materials as they are being developed in order to maximize the probability of their being successful during the implementation of actual training using such materials. Figure 9 is a detailed breakdown of the specific activities that occur during authoring and formative evaluation. The process of formative evaluation planning is illustrated in Figure 11 and shall be carried out in accordance with the models displayed in these two figures and with the steps outlined below.

3.21.1.1 Establish review points and procedures. Review points shall be established to conform to the formative evaluation process outlined in Figure 9. Specific personnel to conduct materials review shall be designated. Provision shall be made for reviews by both content experts (SMEs) and instructional development experts (IDs). Reviews by SMEs shall include validation of all test items. Validation procedures and data interpretation guidelines shall be prescribed. As part of reviews by instructional development experts, individual tryouts shall be conducted. Subjects shall be "naive" representatives of the STC population. Emphasis shall be upon detecting major problems, such as ineffective lesson organization, undefined terms, and incompatible audio and visual components. Procedures for conducting individual tryouts shall be prescribed. Information flow procedures and necessary data forms shall be specified.

3.21.1.2 Determine small group tryout procedures. To organize the subjects for small group tryouts, it is first necessary to determine what the lesson development sequence and schedule is. When the lesson development sequence and schedule is known, a gross schedule for conducting the small group tryouts in terms of numbers of subjects and when they must be available shall be determined. Consideration shall be given to the following:

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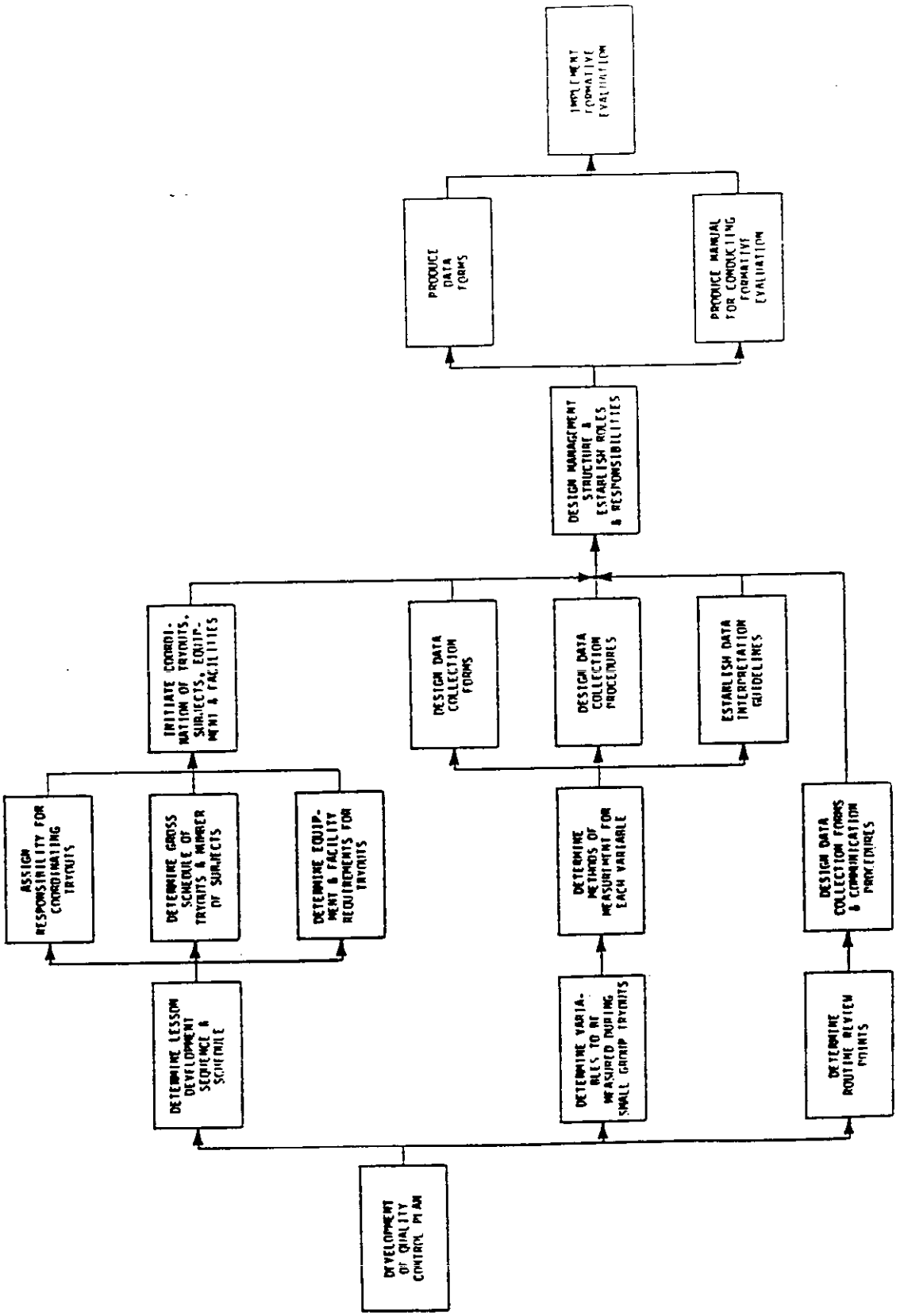


FIGURE 11. Planning for formative evaluation.

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(a) Media used in each STC - Procedures for small group tryouts of lessons in individualized academic media, such as slide-tape, workbook, and videotape, will most likely differ from the procedures used for group academic media (i.e., mediated interactive lectures) and hands-on lessons.

(b) Qualifications of subjects - At a minimum, subjects shall have the same skills and knowledge as students who will enter the STC (e.g., graduate of advanced jet training, graduate of A school, etc.). Preferably, subjects shall be candidates for the ongoing training program (existing systems) or shall be students in the ongoing program who have not received instruction in the subject matter of lessons to be tried out.

(c) Availability of subjects - The number of qualified subjects who will be available for small-scale tryouts shall be determined. When shortages of qualified subjects to try out all lessons are anticipated, a sampling scheme shall be developed. The selection of lessons to be tried out shall be based on subject matter content, type of objectives, and instructional media. For lessons tried out, four or more subjects shall be used to ensure that data is valid.

(d) Continuity of subjects - Prerequisites among lessons and lesson content shall be examined to identify sequences of lessons which should be tried out using the same subjects (i.e., knowledge/skill acquired during a lesson[s] is required for valid evaluation of a succeeding lesson[s]). Scheduling of subjects shall reflect prerequisite requirements.

Parallel to scheduling and specification of subject requirements, any equipment and facility requirements for conducting the tryouts shall also be determined. Responsibility for coordinating small group tryout activities shall be assigned. When these activities are completed, coordination shall be initiated and followed through in order to insure that tryout subjects are available when and where required, with the requisite level of equipment and facility support.

3.21.1.3 Determine variables and methods of measurement for small group tryouts. The main variables shall be learning effectiveness and palatability of the materials to the students. Measurement of instructional effectiveness shall be conducted using test and mastery data. Students shall be questioned directly about the instructional materials in order to determine points where the materials do not communicate well or are difficult to understand or manage. Palatability shall be measured by standard attitude questionnaires which tap the students' basic affective or emotional response to the materials. Once the variables and the methods of measurement for each variable have been determined, the required data collection forms shall be designed and the procedures or methods for collecting this data established. Guidelines for interpreting the data shall be established. Guidelines should assist those who actually implement the small group tryouts in

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interpreting the meaning of data and deciding at what point revisions are necessary. After data collection procedures and interpretation guidelines have been developed, the necessary data forms shall be produced.

3.21.1.4 Determine management structure for formative evaluation. A management structure for the conduct of all formative evaluation procedures shall be established. All review, tryout, and coordination activities for both contractor and government personnel shall be incorporated, along with specific roles and responsibilities. The proposed management structure shall be reviewed with all cognizant personnel to ensure that it is workable within the constraints of the training development environment. A manual shall be produced which details the management structure and roles and responsibilities, containing all review and data collection procedures and forms. This manual shall provide a good working knowledge of the formative evaluation system for those who will participate in the system.

3.21.2 Internal QC planning methodology. Internal QC is the process of trying out finished materials with real students in the real training environment. Its purpose is to identify elements of the instructional materials or the instructional management system that are causing poor learning, inefficiency, or poor student acceptance, and of revising or modifying these elements so as to eliminate the problems they are causing. Planning for internal QC shall cover the time from the first implementation of the newly developed instructional program to the point when training for the weapon system is no longer required. The process of internal QC is shown in Figure 12. The process of planning for internal QC is shown in Figure 13. Planning for internal QC shall be conducted in accordance with the models displayed in these two figures and with the steps outlined below.

3.21.2.1 Determine variables and measurement methods. The basic variables to be measured in internal QC do not differ markedly from those in formative evaluation except that internal QC permits evaluation of training materials in their basic sequence and in the management and administrative context of the training program. The variables to be attended to and the methods of measurement to be used shall be defined in detail. Such variables as learning effectiveness, tests, and student comments shall be employed and shall be supplemented by instructor comments. Student affective factors shall be assessed by standard questionnaires and by data sheets which provide the opportunity for students to comment on the effectiveness of the instructional materials. Instructional system effectiveness shall be measured by comments made by students, instructors, and instructional system managers and administrators.

3.21.2.2 Establish data collection and interpretation procedures. Once the variables and their methods of measurement have been determined, the data collection forms required by the measurement methods shall be designed, and a data collection methodology and schedule shall be established. Simultaneously, as in formative evaluation planning, data interpretation guidelines shall be designed for use by those who will ultimately implement the internal QC system. Data interpretation guidelines shall clearly describe criteria to be used in determining when and what types of revisions are required.

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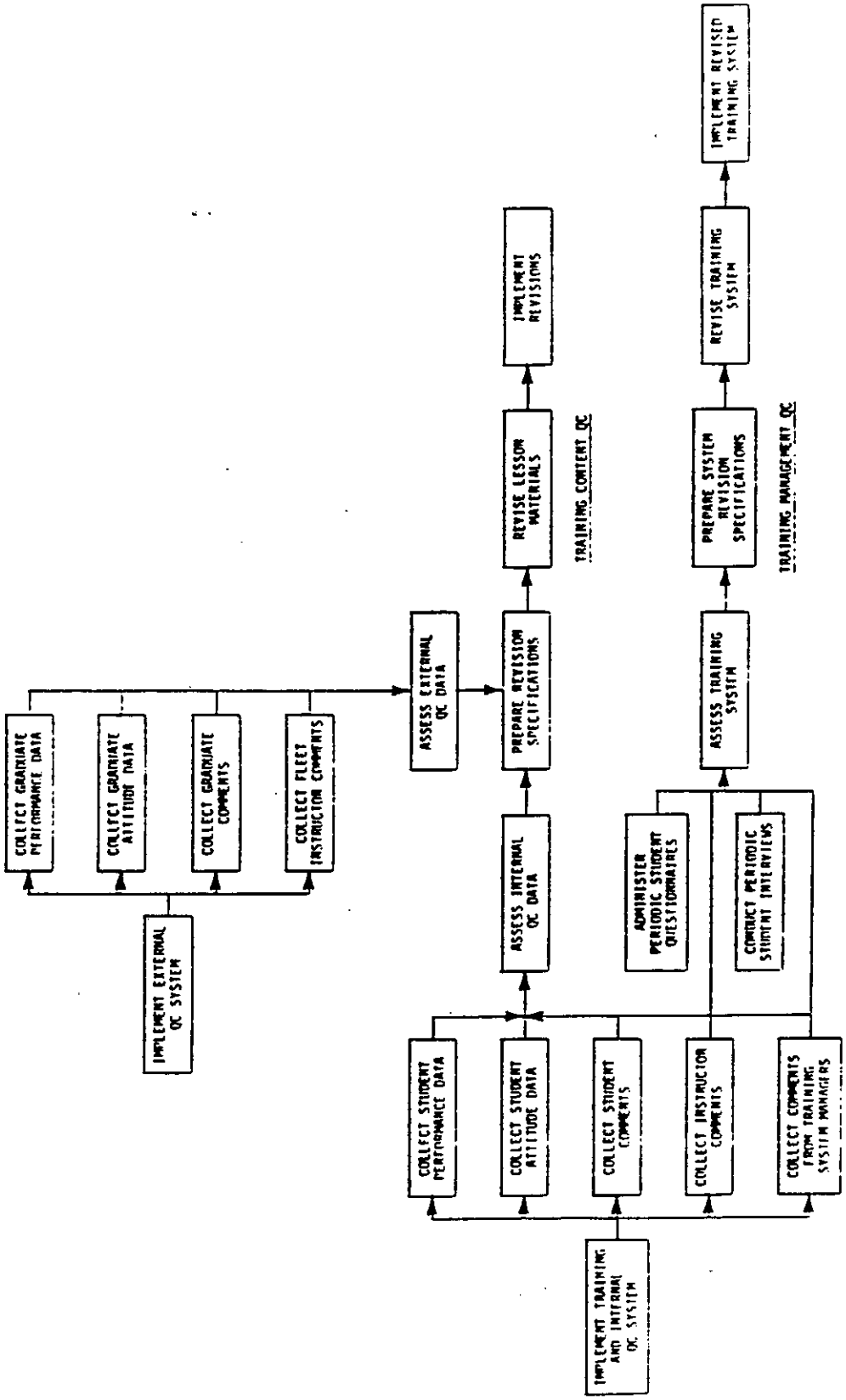


FIGURE 12. The process of conducting internal and external quality control.

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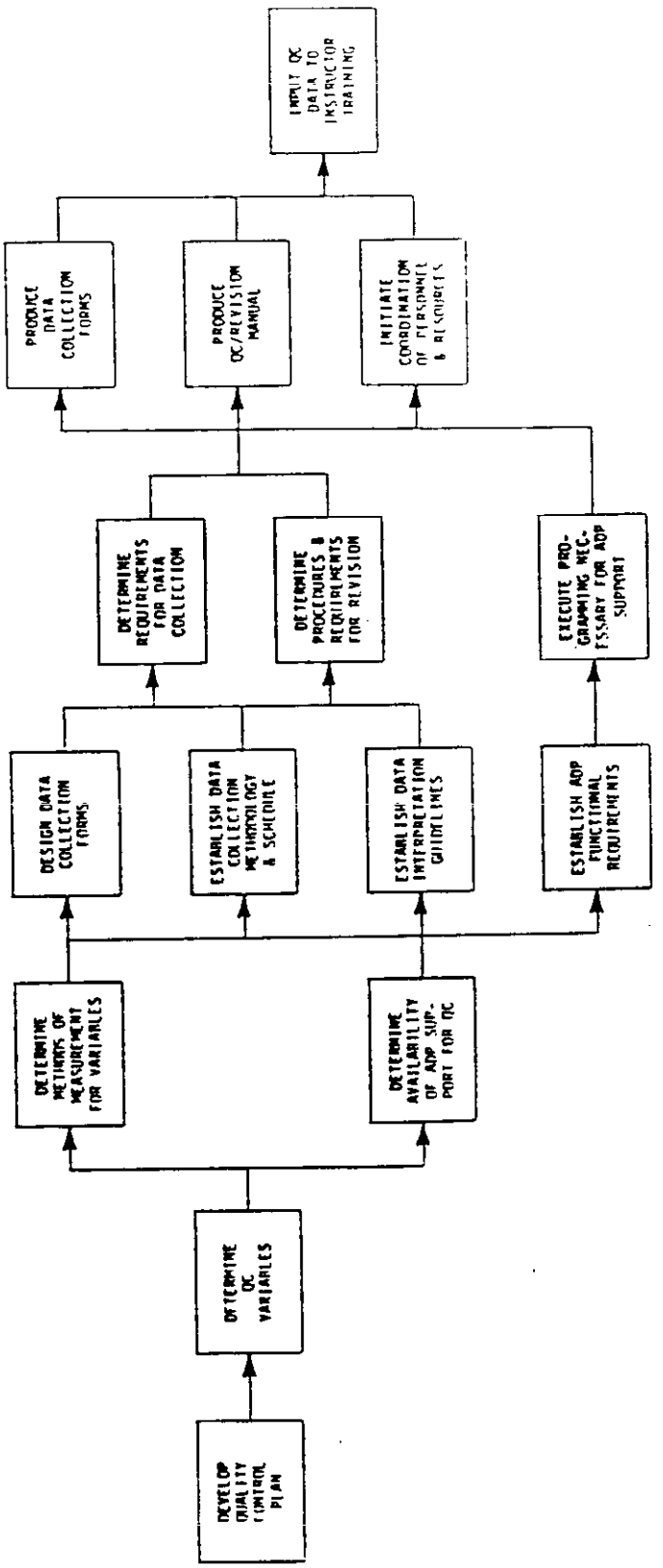


FIGURE 13. Planning for internal and external quality control.

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3.21.3 External QC planning methodology. External QC is the process of evaluating the relevance, currency, and effectiveness of the training system in terms of its ability to meet operational requirements for skilled personnel. The basic model for carrying out external QC shall be the same as shown in Figure 12 for internal QC. Similarly, planning for external QC shall be the same process as shown in Figure 13 for internal QC planning.

3.21.3.1 Determine variables and measurement methods. The variables to be attended to and the methods of measurement to be used shall be defined in detail. The main variables shall be the relevance of the training course subject matter to job requirements in the operational environment and the effectiveness of the course in teaching operational job requirements. External QC shall also determine whether the instruction that was provided in relevant areas was sufficient to produce the level of skill demanded in the operational environment. The external QC system shall pay specific attention to predictable categories of change in content and technique. These include ECPs and modifications to software programs and tactical concepts. The primary data sources in external QC shall be personnel who have graduated from the training program and are now in operational assignments and their supervisors. The basic method of measurement shall be questionnaires/interviews.

3.21.3.2 Establish data collection and interpretation procedures. Once the variables and their methods of measurement have been determined, the data collection forms required by the measurement methods shall be designed, and a data collection methodology and schedule shall be established. Data interpretation guidelines shall be designed for use by those who will ultimately implement the external QC system. Data interpretation guidelines shall clearly describe criteria to be used in determining when and what types of revisions are required.

3.21.4 Determine management structure and resource requirements for internal and external QC and revision. The contractor shall coordinate closely with the training community to develop a management structure for the conduct of all internal and external QC and revision activities. All data collection, interpretation, and production tasks for both contractor and government personnel shall be incorporated, along with specific roles and responsibilities. The management structure shall provide for efficient performance of all tasks and for efficient coordination and communication among all involved personnel. All personnel and other resource requirements (i.e., equipment, facilities, etc.) shall be specified.

3.21.5 Produce data forms and QC manual. Once procedures and requirements for data collection and revision have been determined, the necessary data collection and revision specification forms and a manual detailing all aspects of internal and external QC and revision shall be produced. The QC manual shall be prepared specifically for use by those who will implement and maintain the QC and revision effort. This QC manual shall form a portion of the ITC (3.25).

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3.21.6 SME training for QC plan development. Since development of the QC plan is a contractor activity, no formal SME training shall be required. SME input shall be solicited as required during the development process.

3.21.7 Training of personnel in QC activities.

3.21.7.1 Formative evaluation. In order for SMEs to function adequately during formative evaluation, training shall be provided in the following areas:

(a) Overall structure of the formative evaluation process.

(b) SME roles and responsibilities in the review process.

(1) Content review.

(2) Graphics review.

(3) Use of forms to record review comments.

(c) Individual and small-scale tryout procedures.

(1) Familiarization with the tryout processes.

(2) Monitoring students who are using the instructional materials.

(3) Administering and scoring student tests.

(4) Administering and evaluating student attitude surveys.

(5) Conducting student interviews using previously designed interview questionnaires.

(6) Interpretation of tryout data to identify materials problems.

(7) Development of revision specifications to correct materials problems.

Materials as specified in the appropriate section of UDI-H-25712 shall be developed and used to support SME training.

3.21.7.2 Internal and external QC. Training in internal and external QC procedures shall be provided for training activity and fleet personnel who will be involved in implementing and using the QC plans. This training shall be incorporated in the ITC (3.25) and shall include, but not be limited to, the following:

(a) Familiarization with QC principles and techniques.

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(b) Data collection procedures.

(1) Data entry techniques.

(2) Data input sources.

(3) Data collection points and schedules.

(4) Interview techniques.

(5) Coordination and transmittal of data.

(c) Data interpretation.

(1) Criteria for determining revision requirements.

(2) Specifying required revisions.

(d) Revision.

(1) Interpreting revision specifications.

(2) Development of revised instructional materials.

(3) Modifying the instructional management system.

(4) Front-end analysis, media selection, syllabus sequencing.

3.21.8 QC plan report. Procedures to be used during formative evaluation and internal and external QC shall be documented in accordance with UDI-H-25726 and submitted to the contracting agency for review.

3.22 QC plan update. The QC plan shall be updated at periodic intervals to insure that the most current data is available for use in subsequent ISD steps.

3.22.1 Data sources. The most recent documentation available shall include, but not be limited to, the areas listed below to identify possible additions to, or modifications of, the QC plan:

(a) Media mix data (including complex devices and equipment).

(b) Implementation plan data and updates.

(c) PA data and updates.

(d) TDSP data and updates.

(e) Syllabus and course materials data and updates.

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3.22.2 Methodology. Once it has been determined that a change must be made to a portion of the QC plan, the change shall be effected. Since the nature and scope of update requirements cannot be predicted, it is not possible to specify how the revision process shall be conducted. The general procedures shall be the same as used during development of the initial QC plan. Changes that affect ongoing QC activities shall be implemented immediately.

3.22.3 Training of personnel in updated QC activities. Training in areas affected by the QC plan update shall be provided as per 3.21.7.

3.22.4 QC plan update report. QC plan updates shall be documented as required in accordance with UDI-H-25726 and submitted to the contracting agency for review.

3.23 Implementation plan. A plan for implementing the newly developed instruction in the real training environment shall be developed. This plan shall focus on training management and administration, personnel, procedures, and techniques. It shall be sufficiently detailed and prescriptive to enable training personnel to carry it out without extensive formal training. The implementation plan, along wih the QC plan, shall provide the major input to the development of the ITC (3.25).

3.23.1 Methodology. The process of implementation planning is shown in Figure 14. The implementation plan shall be developed in accordance with this model and as outlined below.

3.23.1.1 Define and describe the proposed instructional system. The principal features of the instructional system shall be based on information concerning the syllabus and the instructional materials. The system shall be characterized in terms of its length, type of sequencing, degree of individualization, degree of self-pacing, type of grading and qualification system, media mix, predominant instructional strategies, and reliance on specialized equipment and techniques.

3.23.1.2 Establish instructional system integration plan and schedule. A plan for integrating/initiating the newly designed training system shall be developed. The plan shall cover procedures for replacing a previous training program or for initiating a wholly new system, perhaps involving a new training site or a unique training organization. The plan shall establish the major events to take place in setting the stage for training implementation. Required activities shall be shown on a detailed time line.

3.23.1.3 Develop equipment storage, maintenance, and utilization plan. This plan shall detail the procedures, schedules, and personnel roles and responsibilities involved in equipment utilization (including scheduling), maintenance, and storage.

3.23.1.4 Develop student grading and evaluation plan. This plan shall establish policy and procedures regarding the grading and qualification of students. It shall specify the test and other data that will be used to

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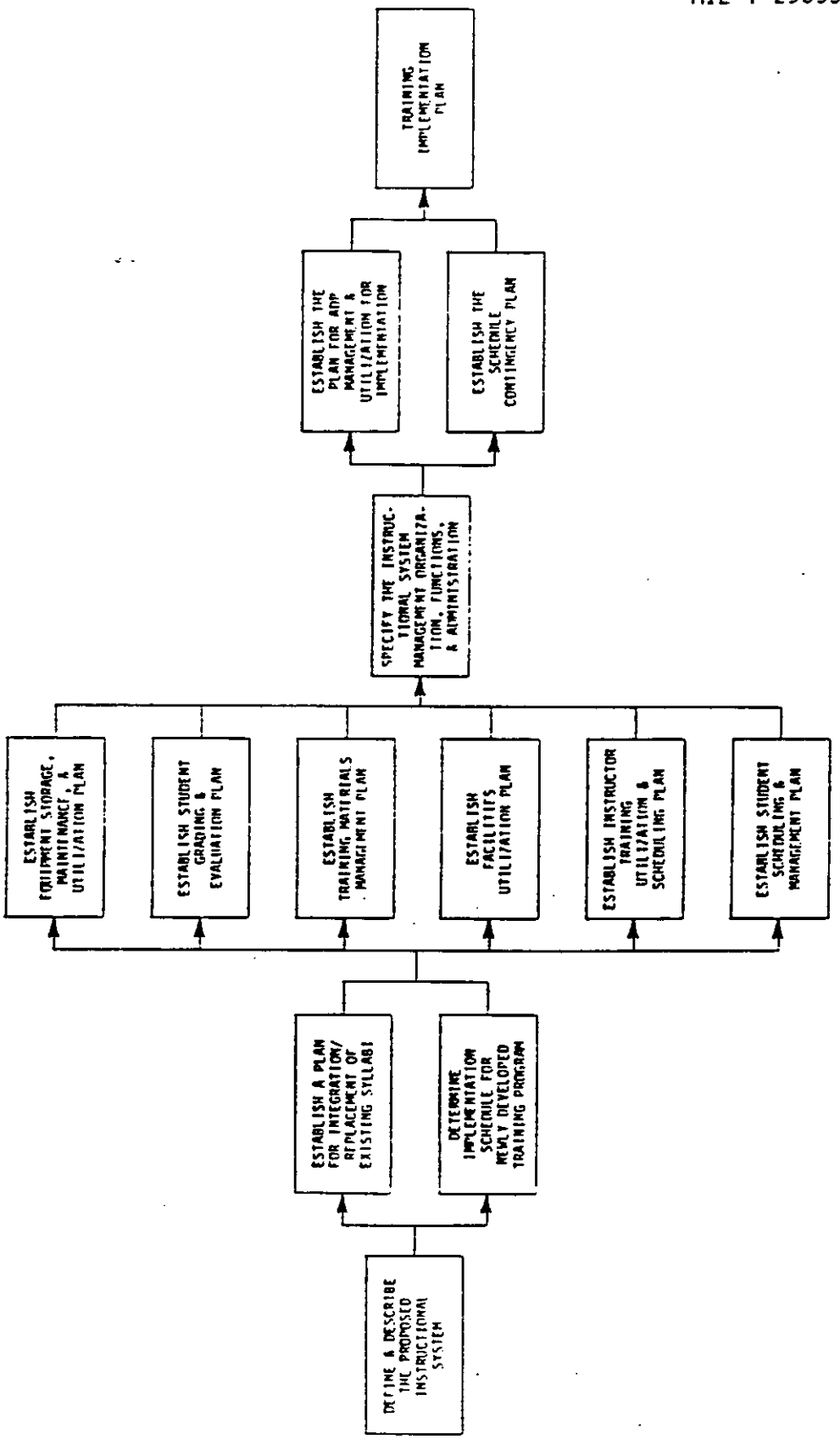


FIGURE 14. Implementation plan development procedure.

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determine pass/fail or other qualification status and how such data will be gathered, stored, processed, and transmitted. The plan shall establish procedures for any type of evaluation contemplated, apart from basic pass/fail determination. The plan shall also address the resources required to implement the plan in terms of personnel, time, workload, equipment, materials, and the like.

3.23.1.5 Develop training materials management plan. This plan shall establish procedures for storage, issue, reproduction, and maintenance of all training materials (e.g., texts, audiovisual programs, manuals and reference materials, etc.). The plan shall provide for controlling and safeguarding materials, as well as for timely, accurate, and flexible materials support for training.

3.23.1.6 Develop facilities utilization plan. This plan shall establish procedures and responsibilities for scheduling, utilization, and maintenance of all training related facilities (e.g., buildings, classrooms, student learning centers [SLCs], etc.).

3.23.1.7 Develop instructor training, utilization, and scheduling plan. This plan shall establish policies and procedures regarding the utilization of instructors within the instructional system. It shall specify their roles and responsibilities with respect to students, materials, syllabus requirements, testing, monitoring, counseling, instructing, and other duties. It shall establish procedures for the training of newly assigned instructors and instructional management personnel. The plan shall establish procedures for scheduling instructors to achieve maximum instructional effectiveness and efficiency.

3.23.1.8 Develop student scheduling and management plan. This plan shall detail the procedures and techniques to be used in scheduling students so that they interface with the syllabus sequence as called for in the course system. The plan shall also establish policies and procedures governing identification and counseling of students with problems and recycling of students through qualification tests and syllabus elements in the event of first time failure.

3.23.1.9 Develop the instructional system management organization, and functions. This part of the implementation plan shall detail the overall instructional systems management organization and functions. It shall identify personnel roles and responsibilities, organizational elements, and communication and coordination channels. It shall detail the functions to be performed by each element in the management structure. The instructional system management organization and functions developed during this step shall provide for efficient, flexible, effective response to systems management requirements as specified in 3.23.1.2 through 3.23.1.8.

3.23.1.10 Develop an automation supplement to the instructional systems management plan. This supplement shall detail procedures and techniques for employing computer assistance in instructional systems management. It shall

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specify personnel roles and responsibilities and describe the management functions and requirements to be computer supported.

3.23.1.11 Develop the schedule contingency plan. This plan shall establish general policies and specific procedures for dealing with events which disrupt training schedules. The plan shall deal in detail with high probability contingencies (i.e., describing courses of action to be taken to minimize the disruption of learning activities).

3.23.2 SME training for implementation plan development. Since development of the implementation plan is a contractor activity, no formal SME training shall be required. Input from SMEs, training administrators, support personnel, and others as required shall be solicited, as appropriate, during the development process. Coordination and communication among all relevant personnel and organizations shall be emphasized to ensure that the implementation plan is workable within the unique policies and constraints of the training environment.

3.23.3 Training of personnel in implementation activities. Training of all personnel who will be involved in implementation shall be a part of instructor training as described in 3.25.

3.23.4 Implementation plan report. Procedures to be used during training program implementation and throughout the life cycle of the training system shall be documented in accordance with UDI-H-25727 and submitted to the contracting agency for review.

3.24 Implementation plan update. The implementation plan shall be updated as required to ensure that the most current data is available for use in training management during implementation of the newly designed training program.

3.24.1 Data sources. The most recent information available shall include, but not be limited to, the areas listed below to identify possible additions or modifications to the implementation plan:

(a) Media mix data (including complex devices and equipment).

(b) QC plan data and updates.

(c) PA data and updates.

(d) TDSP data and updates.

(e) Syllabus and course materials data and updates.

3.24.2 Methodology. Once it has been determined that a change must be made to a portion of the implementation plan, the change shall be effected. Since the nature and scope of the update requirements cannot be predicted, it is not possible to specify how the revision process shall be conducted. The general procedures shall be the same as used during development of the

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initial implementation plan. Changes that affect ongoing implementation activities shall be implemented immediately. Necessary coordination of implementation plan changes with QC activities shall be ensured.

3.24.3 SME training for implementation plan update. No formal SME training shall be required. SME input shall be solicited as necessary during the update process.

3.24.4 Training of personnel in updated implementation activities. Training shall be provided as required to ensure that all affected personnel are conversant with the revised implementation activities.

3.25 Instructional system operation. Procedures and materials for training instructors to perform their duties in the training system shall be developed. The course shall address all aspects of the training system, with particular emphasis on instructor roles and responsibilities.

3.25.1 Training system components, features, and operating procedures. In order for training personnel to properly use the instructional system which is being developed for them, they shall be provided with training in the following areas:

(a) Introduction to the training organization.

(1) Training organization personnel, relationships, roles, communication channels, etc.

(2) Training organization policies.

(3) Role of support elements and relationships to the training organization.

(4) Summary of student characteristics.

(b) Description of the training program.

(1) Overview of instructional strategies.

(2) Syllabus organization and student flow.

(3) Roles of ISD team.

(4) Introduction to media.

(c) Training system operation and scheduling.

(1) Overview of instructor roles and responsibilities.

(2) SLC operating procedures.

(3) Scheduling SLC resources.

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(4) Student scheduling.

(5) Instructor scheduling.

(d) Instructor roles in the learning center.

(1) Evaluation of tests.

(2) Student remediation.

(3) Responding to student questions/problems.

(4) Recording test scores.

(5) Check-in and out of materials.

(e) Conducting lectures and discussion sessions.

(1) Use of lecture guides.

(2) Testing.

(3) Use of visuals.

(4) Use of student guides.

(5) Student-instructor interaction.

(6) Student remediation.

(f) Conducting trainer and operational equipment exercises.

(1) Session formats.

(2) Use of scenario guides.

(3) Use of performance checklists.

(4) Instructional techniques.

(5) Briefing and debriefing procedures.

(6) Student remediation.

(g) Record keeping.

(1) Record and report requirements.

(2) Instructor record keeping responsibilities.

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(3) Monitoring student progress.

(4) Identifying problem students.

(h) QC and revisions.

(1) Overview of QC system.

(2) Sources of QC data.

(3) Interpretation of QC data.

(4) Revision procedures.

(i) Storing, operating, and troubleshooting media devices.

(1) Instructor versus maintenance responsibilities.

(2) Preventive maintenance.

(3) Normal operating procedures.

3.25.2 SME training for the development of training system materials. Since development of the training system materials is a contractor activity, no formal SME training shall be required. SME input, however, is important to the development process and shall be solicited as required.

3.25.3 Training instructors in training system operation. Prior to implementation of the training course(s), training of the initial cadre of instructors shall be accomplished. This instruction shall ensure that all relevant squadron personnel are well versed in the operation of the system and that roles and responsibilities are clearly conveyed. Subsequent training of replacement instructors shall be accomplished by appropriate personnel using individualized materials.

3.25.4 ITC materials. Materials to be used to train instructors in operation of the training system shall be documented in accordance with UDI-H-25728 and submitted to the contracting agency for review.

3.26 Training device operation. Procedures and materials for training instructors to operate the training devices shall be developed. The course will address specific trainer operation from an instructor/operator aspect only. It will not include such subjects as trainer maintenance and curriculum, scenario, or training mission profiles developed during the ISD process.

3.26.1 Trainer capabilities and characteristics. General trainer characteristics, capabilities, equipment, systems, and subsystems shall be presented. The description shall contain a listing of the objectives which are practiced in the trainer.

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3.26.2 Student station(s). A detailed review of all station controls, instrumentation, and other displays, including additional environmental capabilities relative to motion, visual, radar, infrared, landmass, navigational, and other cues and feedback, shall be accomplished. Special attention shall be devoted to operator/student cues or functions that are constrained or cannot be performed realistically.

3.26.3 Trainer operation. Academic and hands-on practice training shall be accomplished to provide a working-level understanding of all applicable displays, instructor cues, aids, problem or scenario controls, preprogrammed maneuver requirements, and instructor/operator feedback provisions. The following applicable features of the instructor/operator station shall be presented and reviewed to ensure working-level knowledges and skills have been mastered.

(a) Problem control capabilities, including, but not limited to, control of independent and dependent variables: freeze, replay, voice transmissions, performance rate/pacing, target control, and system capability control.

(b) Location and function of displays and controls; input requirements, including content, format, symbology, etc.

(c) Instructional management features not automatically provided for by software. Examples would include features relative to initial conditions, initialization, monitoring student performance, malfunction insertions, environmental status or system status, such as television monitors or panel instrumentation displays.

(d) Programmable or automated instructional management features. Examples would include problem initialization, scenario or maneuver sequencing, performance measurement, scoring, malfunction insertion, demonstrations, and incorrect response indications.

Instruction in use of instructional materials produced for trainer lessons (i.e., instructor guides, evaluation checklists, etc.) shall be provided, and practice scenarios using sample trainer lessons from the operator or maintenance personnel course under development shall be used during hands-on practice.

3.26.4 SME training for the development of training device materials. Since development of the training device operation materials is a contractor activity, no formal SME training shall be required. SME input, however, is important to the development process and shall be solicited as required.

3.26.5 Training instructors/operators in training device operation. Prior to implementation of the training course(s), training of the initial cadre of training device instructors/operators shall be accomplished. Academic and hands-on practice sessions shall be conducted and shall be

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supported by appropriate audiovisual aids. Subsequent training of replacement instructors shall be accomplished by appropriate training activity personnel using individualized materials.

3.26.6 Training device instructor/operator training materials. Materials to be used to train device instructors/operators shall be documented in accordance with UDI-H-25729 and submitted to the contracting agency for review.

3.27 Student orientation. Procedures and materials for training students to function in the training system shall be developed. The instruction shall address all aspects of the training system with particular emphasis on student roles and responsibilities and day-to-day activities.

3.27.1 Training system components, features, and operating procedures. Most of the content of the student orientation shall be drawn from the ITC. Elements of the ITC shall be modified as required for student orientation. The instruction shall be developed specifically for student use and shall contain only information necessary to enhance student understanding of, and performance in, the training environment. Topic areas covered shall include, but not be limited to:

(a) Introduction to the training organization.

(1) Training organization policies.

(2) Training organization personnel, relationships, roles, communication channels, etc.

(3) Student interaction requirements, communication channels, etc.

(b) Description of the training program.

(1) Length.

(2) Introduction to media.

(3) Syllabus organization.

(4) General characteristics (e.g., pacing, evaluation, student progress).

(c) Training system operation.

(1) Overview of instructor roles and responsibilities.

(2) SLC operating procedures.

(3) Student scheduling.

(4) Student responsibilities.

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(d) Use of instructional materials, equipment, and facilities.

(1) Guidelines for preparing for lessons in each medium.

(2) Components of a lesson in each medium and how to use them.

(3) Facilities locations, hours of operation, secure areas, etc.

3.27.2 SME training for the development of student orientation materials. Since development of the student orientation materials is a contractor activity, not formal SME training shall be required. SME input shall be solicited as required during the development process.

3.27.3 Conducting student orientation. Prior to the formal start of training for each class in weapon system operation or maintenance, the student orientation shall be conducted. The instruction shall insure that all students are well versed in the operation of the system and in their roles and responsibilities.

3.27.4 Student orientation manual. Materials to be used to train students in operation of the training system shall be documented in accordance with UDI-H-25730 and submitted to the contracting agency for review.

3.28 Administration and management. Administration and management tasks shall be carried out to ensure that program activities are properly planned, reported while in progress and as completed, and summarized at program completion. These requirements are described in the following paragraphs.

3.28.1 Program work plan. The program work plan shall provide the information and structure against which the government can monitor program progress during the period of performance. It shall provide the basis for program execution and shall contain clearly verifiable milestones.

3.28.1.1 Methodology. The work plan shall be based upon relevant sections of the contractor's proposal. Staffing, scheduling, and management areas shall be modified as necessary to reflect changes resulting from preaward negotiations and to meet the following requirements:

(a) Management functions shall be clearly delineated. They shall present a management structure and plan whereby all contract requirements can be met on time and within budget, and through which strict QC standards can be implemented.

(b) Assignment of personnel by name, labor category, and estimated man-hours per contract task shall be presented. The man-hour estimates shall clearly demonstrate an efficient assignment of personnel to meet contract requirements.

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(c) A detailed contract schedule shall be developed. Milestones shall be clearly indicated. There shall be no more than one month between contract milestones.

3.28.1.2 Work plan report. The program work plan shall be documented in accordance with UDI-H-25731 and submitted to the contracting agency for review.

3.28.2 Program progress reporting. Periodic progress reports shall provide the government with information necessary to monitor program progress. Progress shall be reported relative to the schedule, milestones, and man-hour estimates contained in the work plan.

3.28.2.1 Methodology. The progress reports shall be systematically developed so that the program status is clearly and concisely defined in each report. Each of the following topic areas shall be discussed:

(a) Status of active tasks and a summary of tasks completed during the reporting period.

(b) Problems encountered during the reporting period with recommended or implemented solutions.

(c) Visits and meetings which occurred during the reporting period and the purpose and results of each.

(d) Tasks scheduled to be in progress or completed during the next reporting period.

(e) Resources expended during the reporting period and cumulative expenditures since the start of the contract.

3.28.2.2 Progress reports. Program progress shall be documented in accordance with UDI-H-25772 and submitted to the contracting agency for review.

3.28.3 QC during production and implementation. All QC activities and validation data concerned with the production and implementation of materials shall be summarized. These activities and data shall clearly indicate the types of problems encountered during formative evaluation and internal and external QC and the actions and revisions required to correct the problems.

3.28.3.1 Methodology. Formative QC activities and data shall be based upon procedures outlined in the QC plan and on data summarized in progress reports submitted during the production phase. Internal and external QC activities and data shall be based upon QC plan procedures and data summarized in progress reports submitted during the implementation phase. At a minimum, the following information and data shall be presented:

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(a) Formative evaluation and internal and external QC procedures, with particular emphasis on problems, solutions, and recommendations to prevent future problems and any modifications to procedures presented in the QC plan, along with rationale for the modifications.

(b) Descriptions of all formats and statistics used to report the data.

(c) Formative evaluation and internal and external QC data for each lesson in each course.

(d) Summary data across each course.

(e) Data interpretation guidelines for developing revision specifications.

3.28.3.2 QC report. QC activities and data shall be documented in accordance with UDI-H-25773 and submitted to the contracting agency for review.

3.28.4 Program summary. Upon completion of all program tasks, summary information to document the total effort shall be presented. This information shall be presented in report format in accordance with MIL-STD-847, prepared by or for the Department of Defense.

3.28.4.1 Methodology. A complete accounting of all program activities, schedules, and end products shall be provided. Primary sources of information shall be the end products/reports submitted throughout the program. Topic areas shall include, but not be limited to, the following:

(a) Background information to place the program in the proper historical perspective.

(b) A summary of all program activities to include program organization, schedules, and specific tasks carried out.

(c) Complete discussion of problems encountered, solutions, and recommendations to prevent problems in future programs.

(d) Results of the program with emphasis on instructional characteristics, statistical summaries (e.g., syllabus length, number of objectives, etc.), and evaluation during implementation.

(e) Conclusions and recommendations with emphasis on improvements in ISD procedures, insights gained, and general observations.

(f) A complete summary of resources expended to include man-hours, materials, travel, equipment, and others as appropriate.

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(g) Samples of interim products which trace ISD development steps (i.e., task listings, objectives hierarchies, media selection and syllabus, lesson specifications, and lesson materials and tests).

3.28.4.2 Training program work report. The summary of program activities shall be documented in accordance with UDI-H-25774 and submitted to the contracting agency for review.

4. QUALITY ASSURANCE (QA) PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 QA provisions, general. An effective QA program shall be organized and applied to the specified processes described in Section 3 herein. The application of QA practices shall apply whether all or part of the ISD process is specified.

4.2.1 As a minimum QA practice, the contractor shall conduct internal design reviews to ensure technical accuracy and completeness of work, as well as to ensure the integration of all aspects of the ISD process. Other QA practices shall provide for periodic user (fleet) input and review to ensure a responsive and useful training program.

4.2.2 The QA program shall be imbedded in both technical and managerial aspects of all projects where this specification is applied.

4.2.3 Acceptance criteria. For the instruction to be accepted, on the first attempt at least 85 percent of the students shall meet the standards established for each objective.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the QC plan and in Sections H and I of the Contract Schedule.

4.4 Records. Records of all raw data, decisions, analyses, and other data which affect the quality or QA aspects of the ISD process shall be preserved and submitted to the procuring agency at the end of the project.

5. PACKAGING

5.1 All deliverable items shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct

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shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations. Additional requirements shall be as specified in Schedule G of the Contract Schedule.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The Military Specification, Requirements for Training System Development (MIL-T-29053B), will be used to prescribe the procedures to be followed in an ISD program. Procedures are applicable to programs in which all ISD steps are sequentially executed and to programs in which selected steps in the ISD process are prescribed to solve specific training problems.

6.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified in Section 3 will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.1.2	Problem Analysis Report	UDI-H-25710
3.2.2	Training Development and Support Plan Report	UDI-H-25711
3.4.1.3, 3.4.2.2, 3.5.1.3, 3.5.2.2, 3.8.3, 3.12.6, 3.17.4, 3.20.2, 3.20.3	Subject Matter Expert Training Materials	UDI-H-25712
3.4.3, 3.5.3	Task Listings Report	UDI-H-25713
3.7.3	Student Entry Level Report	UDI-H-25714
3.8.4	Objectives Hierarchies Report	UDI-H-25715
3.10.6	Training Equipment Survey Report	UDI-H-25716
3.11.1	Media Selection Model Report	UDI-H-25717
3.11.3	Trainer Functional Description Report	UDI-H-25718
3.11.4	Trainer Modification Report	UDI-H-25719
3.12.7	Media Selection and Syllabus Report	UDI-H-25720
3.15.3	Training Support Requirements Report	UDI-H-25721
3.17.5	Lesson Specification Report	UDI-H-25722
3.19.2	Lesson Format Guides	UDI-H-25723
3.20.4	Student Training Materials	UDI-H-25724
3.20.5	Training Materials Development Report	UDI-H-25725
3.21.8	Quality Control Plan Report	UDI-H-25726
3.23.4	Implementation Plan Report	UDI-H-25727
3.25.4	Instructor Training Course Materials	UDI-H-25728
3.26.6	Training Device Instructor/Operator Training Materials	UDI-H-25729

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Paragraph	Data Requirement	Applicable DID
3.27.4	Student Orientation Manual	UDI-H-25730
3.28.1.2	Work Plan Report	UDI-H-25731
3.28.2.2	Progress Reports	UDI-H-25772
3.28.3.2	Quality Control Report	UDI-H-25773
3.28.4.2	Training Program Work Report	UDI-H-25774

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Miscellaneous notes.

6.3.1 Reference documents. The following documents may be useful in explaining some specific aspects of the ISD process:

(a) AFM 50-2: "Instructional System Development."

(b) AFM 50-58: "Handbook for Designers of Instructional Systems."

(c) NAVTRA 106A or TRADOC PAM 350-30: "Interservice Procedures for Instructional Systems Development."

(d) CNTT-A10: "Procedures for the Planning, Design, Development, and Management of Navy Technical Training Courses."

(e) NAVEDTRA 110: "Procedures for Instructional Systems Development."

6.3.2 Sample procedure for development of maintenance task listing. If selected, the method described below will satisfy the requirements of 3.5.1.2 of this specification.

6.3.2.1 Define areas of maintenance responsibility. The boundaries and depth of the maintenance domain shall be established. The pieces of equipment that must be covered and which may be excluded and the level of detail maintenance that must be applied in each equipment item shall be defined. These factors shall be isolated in the following way:

(a) Conduct an analysis and enter data in a matrix such as that shown in Figure 15 for the following:

(1) In the category column, consider operational equipment, automatic test equipment, training devices, and other system maintenance support equipment.

(2) In the area column, for each of the above categories, consider area of hardware and software of the following levels of detail: system, unit, and sub-unit.

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CATEGORY	AREA	TYPE AND LOCUS OF ACTION			
		MAINT/OPER	LOCUS	MAINT/OPER	LOCUS
OPERATIONAL EQUIPMENT	(a) Hardware (System)	M	ORG. LVL.	---	---
	(b) Hardware (Unit)	M	ORG. LVL.	M	INT. LVL.
	(c) Hardware (Sub-unit)	M	ORG. LVL.	M	INT. LVL.
	(d) Software (for System Test)	O	ORG. LVL.	M	INT. LVL.
AUTOMATIC TEST EQUIPMENT	(e) Hardware (for Unit Test)	O	INT. LVL.	M	INT. LVL.
	(e) Software (for Unit Test)		INT. LVL.	M	INT. LVL.
	(e) Hardware (for Sub-unit)	O	INT. LVL.	M	INT. LVL.
	(e) Software (for Sub-unit)		INT. LVL.	M	INT. LVL.
	(f) Hardware (for Unit Tester Test)	O	INT. LVL.	M	INT. LVL.
	(f) Software (for Unit Tester Test)		INT. LVL.	M	INT. LVL.
	(f) Hardware (for Sub-unit Tester Test)	O	INT. LVL.	M	INT. LVL.
	(f) Software (for Sub-unit Tester Test)		INT. LVL.	M	INT. LVL.
TRAINING DEVICES	(g) Hardware (Devices)	M	TRNG. AREA	---	---
	(h) Hardware (Unit)	M	TRNG. AREA	M	INT. LVL.
	(i) Software (for Device Test)	O	TRNG. AREA	M	INT. LVL.
OTHER SYSTEM MAINTENANCE SUPPORT EQUIPMENT	(j) Hardware (Equipment)	O/M	ORG. LVL.	---	---
	(k) Hardware (Unit)	M	ORG. LVL.	M	INT. LVL.

FIGURE 15. Areas of maintenance responsibility.

(a) Examples: Fuel system, propulsion system, flight control system.
(b) Typically a black box or other removable assembly.
(c) Typically a card within a box, or a part of a mechanical unit.
(d) Built-in test provisions at the system level.
(e) Test sets and computer programs for testing boxes and cards.
(f) Test set provisions for testing the testers.
(g) Examples: Flight simulator, weapon system simulator, hot plant simulator, system simulator.
(h) Typically a black box or other simulator - removable assembly.
(i) Built-in test provisions at the simulator system level.
(j) Specific pieces of other equipment used to support the system.
(k) Typically a black box or other system maintenance support system - removable assembly.

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(3) In the type and locus columns, for each of the above areas, consider the type and locus of action required, in terms of maintenance (M) or operation (O), and work location.

(4) Make cell entries as shown but geared to the weapon system under examination. Where no maintenance or operation requirement exists, enter "---" in the applicable cell.

(b) Determine the scope of maintenance responsibility through analysis of the completed matrix.

6.3.2.2 Identify required maintenance tasks. The entire population of maintenance tasks shall be determined. This shall be accomplished in accordance with the following steps. For each item in the category column in Figure 15, conduct an analysis and enter data in a matrix such as that shown in Figure 16 for the following:

(a) Step 1. In the nomenclature column, list units per weapon system. Then continue, listing sub-units per unit. Finally, identify the software program(s) at the system test level.

(b) Step 2. Assign numbers reflecting item status in the equipment hierarchy.

(c) Step 3. In the maintenance function column, note that all aspects of maintenance work are covered, including operator.

(d) Step 4. For each item in the nomenclature column, identify all applicable maintenance requirements. Whenever a requirement occurs, mark the corresponding cell with a small "x" in the upper left corner, leaving room for another symbol later. Where no maintenance requirement exists, leave the cell blank.

(e) Step 5. Label each sheet to show the personnel type to be assigned to the work and the applicable work area (i.e., organizational level, intermediate level, etc.).

(f) Step 6. Go back through the matrix and place a maintenance task number in each occupied cell. For each personnel type, start with "1" and number consecutively.

(g) Step 7. Use the completed matrix as an indication of the specific maintenance tasks required on system equipment.

(h) Step 8. Repeat (a) through (c) for automatic test equipment, using the matrix shown in Figure 17.

(i) Step 9. Repeat (a) through (g) for training devices, using the matrix shown in Figure 18.

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SPECIFIC MAINTENANCE TASKS

OPERATIONAL EQUIPMENT				MAINTENANCE FUNCTION								
				A	B	C	D	E	F	G	H	I
PERSONNEL TYPE	WORK AREA	NUMBER	NOMENCLATURE	CHECKOUT	INSPECT	TROUBLESHOOT	SERVICE	REPAIR IN PLACE	REMOVE/INSTALL	DISASSEMBLE OR ASSEMBLE	CALIBRATE	OPERATE
		1.0	First System									
		1.1 1.2	First Unit in First System Second Unit in First System Etc.									
		2.0 2.1	Second System First Unit in Second System Etc.									
		1.2 1.2.1 1.2.2	Second Unit in First System First Sub-unit in Second Unit Second Sub-unit in Second Unit Etc.									
		1.P	Software Program for First System Test									

FIGURE 16. Suggested method of specifying maintenance tasks on operational equipment.

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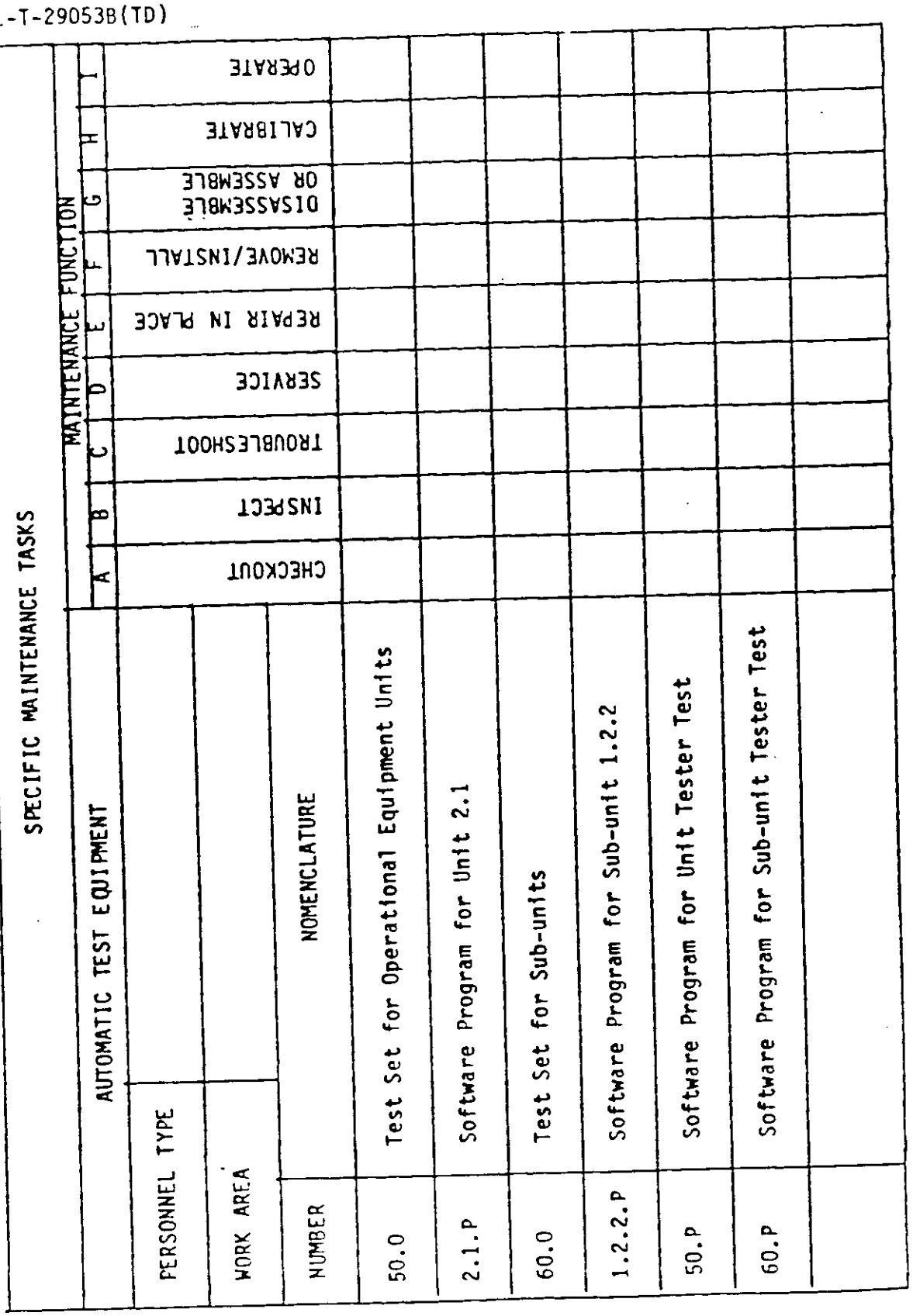


FIGURE 17. Suggested method of specifying maintenance tasks on automatic test equipment.

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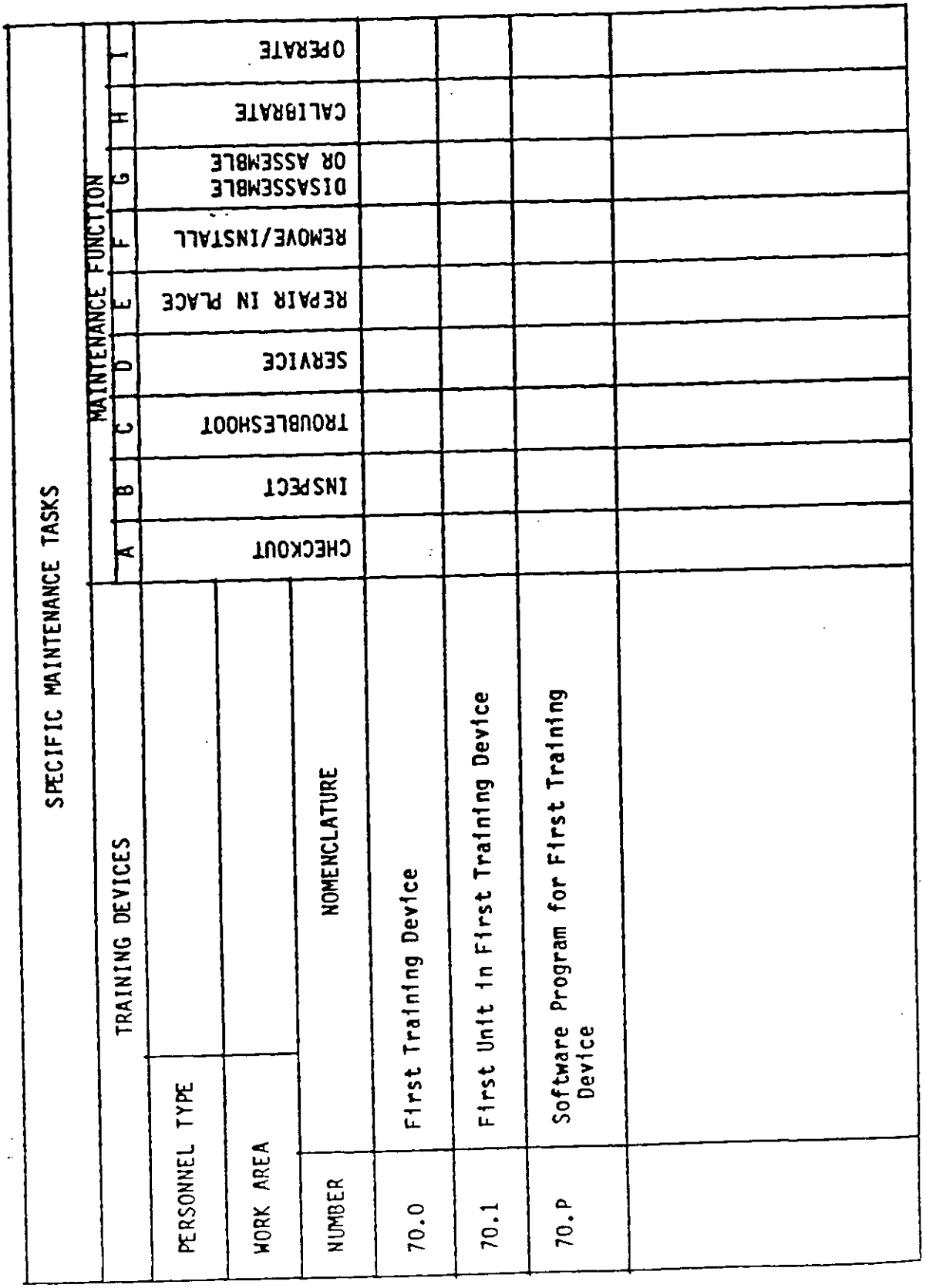


FIGURE 18. Suggested method of specifying maintenance tasks on training devices.

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(j) Step 10. Repeat (a) through (g) for other maintenance and support equipment, using the matrix shown in Figure 19.

6.3.2.3 Prepare maintenance task listings. This operation is to identify and define job performance requirements and organize them into their proper relationship with each maintenance task in such a way as to enable training decisions later in the ISD process. Job performance requirements shall contain the following three parts: behavioral statement, conditions of performance, and performance acceptance criteria. To achieve these objectives, the following procedure may be used:

(a) Prepare a form such as that shown in Figure 20. Provide space for personnel type and for processing controls as necessary. Provide space for listing maintenance tasks, standards, and conditions. Complete the form for each personnel type.

(b) Obtain maintenance task data from the matrix series generated in 6.3.2.2. Maintenance task numbers will be as shown sequentially in the matrix cells. Each maintenance function will be expressed alphabetically as shown above the cells. Each equipment will be expressed by decimal number as shown preceding nomenclature on the matrix. When encoded in this manner, and used in conjunction with relevant matrix records, such entries will constitute an acceptable set of behavioral statements.

(c) Enter standards data at least in terms of speed and accuracy.

(d) Enter conditions data at least in terms of the variables shown. Employ the following types of entries:

(1) Complexity of activity - high, medium, low.

(2) Frequency of activity - high, medium, low.

(3) Psychomotor skills required - yes, no.

(4) Hands-eyes available (for reading job aid) - yes, no.

(5) Job aid permitted - yes, no (see [e] below).

(6) Job aid exists - yes, no.

(7) Environment - dark, light, noise, space, etc.

(8) Personnel - alone, part of team, under supervision.

(9) Equipment - available tools or equipment.

(10) Problem definition - stimulus present to initiate the task.

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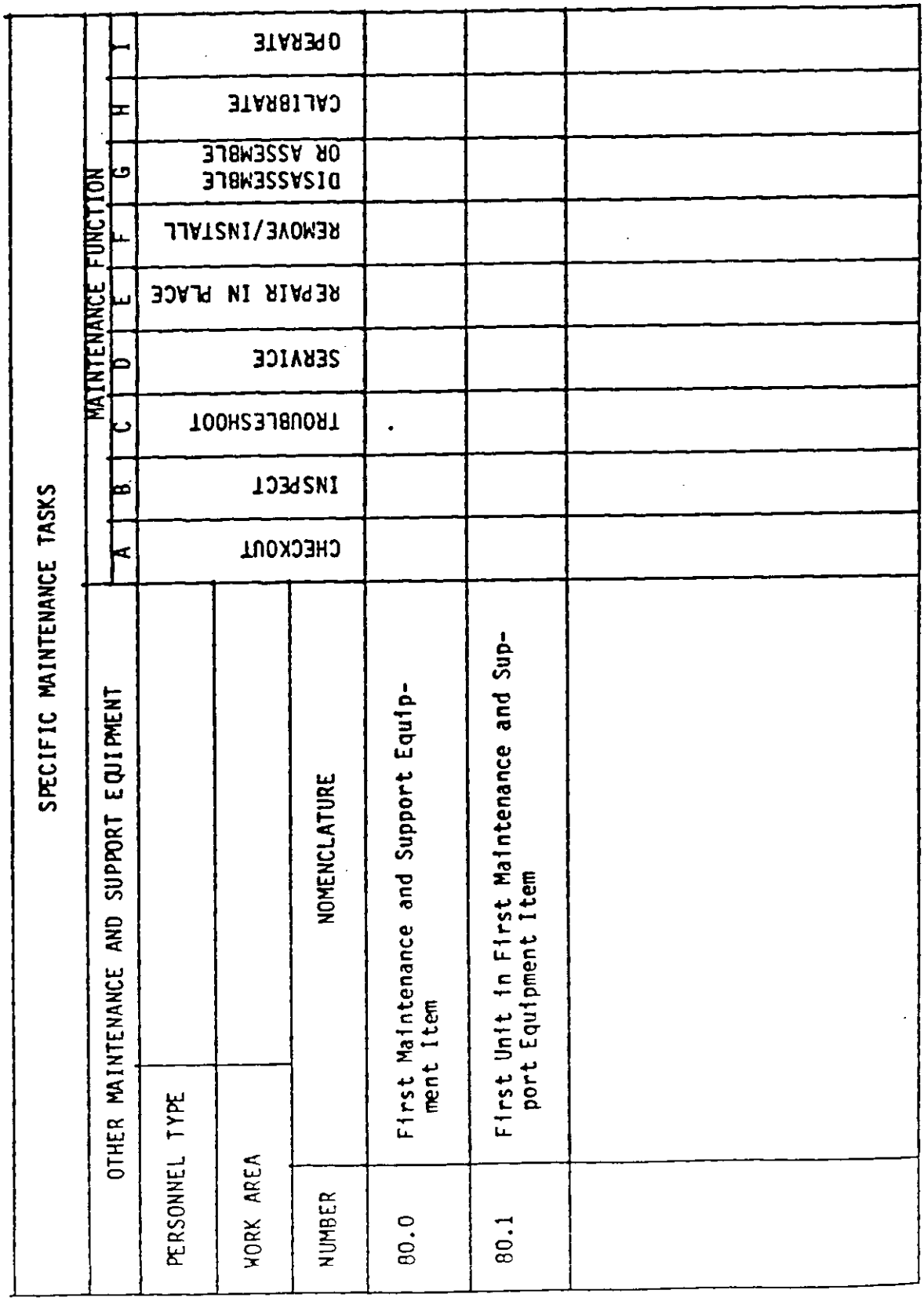


FIGURE 19. Suggested method of specifying maintenance tasks on other maintenance and support equipment.

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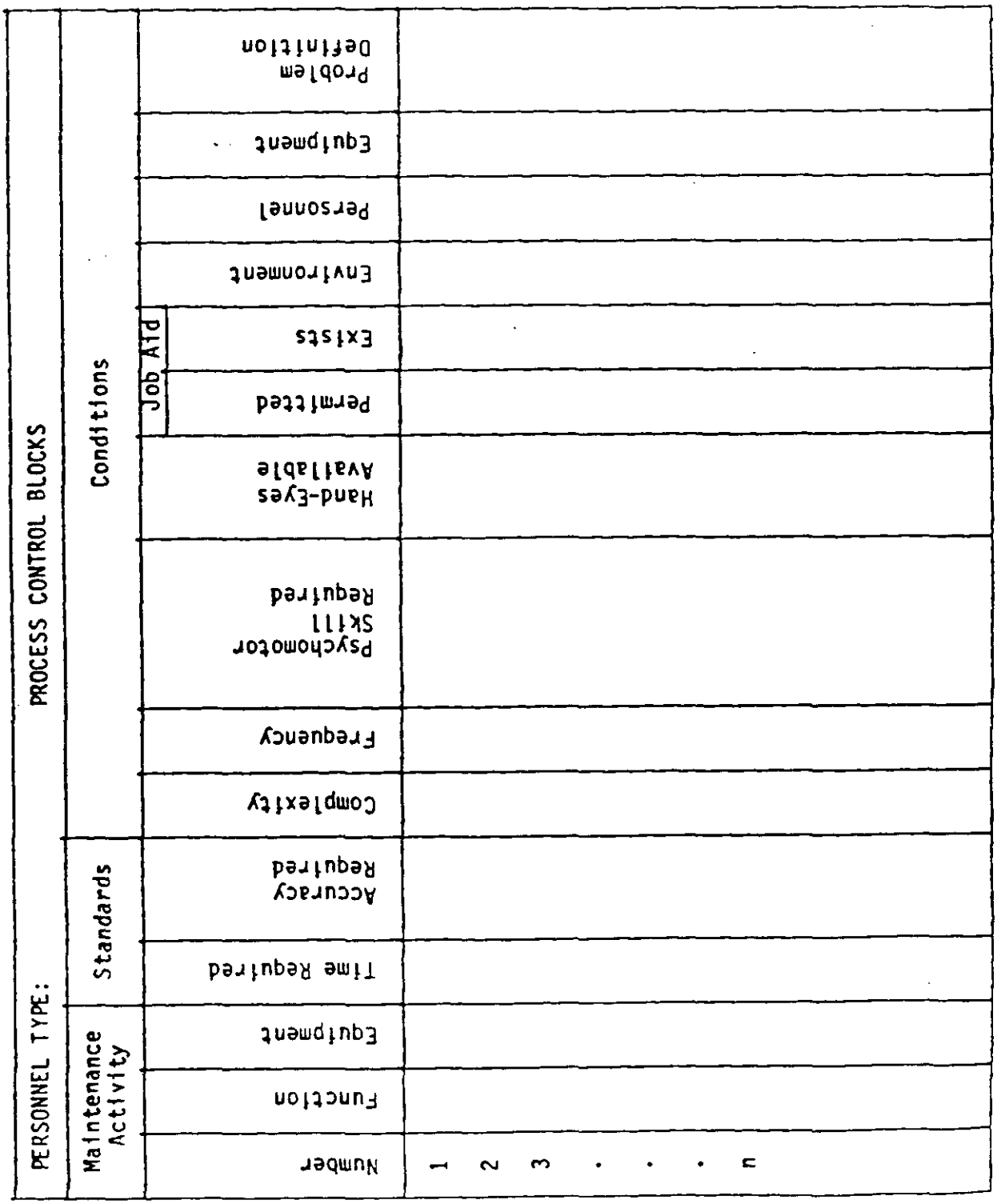


FIGURE 20. Maintenance task listing format.

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(e) Use the standards and conditions data to determine whether or not a job aid, such as printed directions, should be permitted in lieu of task training. A decision model for this purpose is presented in Figure 21.

Decision Model for Job Aid vs. Training

1. Make job aids available for entry-level personnel on every required maintenance activity except:

   (a) Troubleshooting.

   (b) Activities demanding psychomotor skills.

   (c) Activities denying hand-eye access to job aid during work period.

   (d) Task activities requiring high speed performance.

2. Require personnel to use applicable job aids until they demonstrate their ability to perform effectively without such aids.

3. Notwithstanding Rule 2, require personnel to use applicable job aids on all activities of high complexity and low frequency.

FIGURE 21. Decision model for job aid vs. training.

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INDEX

Paragraph		Page
3.17.1	Academic lesson specification development.	44
3.11.5	Academic media selection	31
4.2.3	Acceptance criteria.	80
3.28	Administration and management.	77
3.7	Analysis of entry level behavior	17
2	APPLICABLE DOCUMENT.	2
3.10.2	Associated training equipment.	26
3.27.3	Conducting student orientation	77
3.12.5	Daily class schedule	38
6.2	Data requirements.	81
	Data sources	
3.24.1	Implementation plan update	71
3.18.1	Lesson specification update.	53
3.5.1.1	Maintenance task listings development.	14
3.8.1	Objectives hierarchies	18
3.9.1	Objectives hierarchies update.	24
3.6.1	Operator and maintenance task listings update.	17
3.4.1.1	Operator task listings development	7
3.22.1	Quality control plan update.	67
3.14.1	Syllabi update	40
3.12.4	Determination of media plans	38
	Determination of training tracks	
3.13.1	Maintenance syllabi.	39
3.12.1	Operator syllabi	32
3.21.4	Determine management structure and resource requirements for internal and external quality control and revision	65
3.17.2	Hands-on lesson specification development.	50
3.11.2	Hands-on media selection	29
3.23	Implementation plan.	68
3.23.4	Implementation plan report	71
3.24	Implementation plan update	71
4.3	Inspection conditions.	80
3.25	Instructional system operation	72
3.25.4	Instructor training course materials	74
6.1	Intended use	81
2.1	Issue of documents	2
3.19.2	Lesson format guides	54
3.19	Lesson formatting.	53
3.20	Lesson materials and tests	54

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:25Z
Check the source to verify that this is the current version before use.

INDEX
(Continued)

Paragraph		Page
3.17.5	Lesson specification report.	52
3.18	Lesson specification update.	52
3.18.3	Lesson specification update report	53
3.13	Maintenance syllabi.	39
3.5	Maintenance task listings.	14
3.5.1	Maintenance task listings development.	14
3.5.3	Maintenance task listings report	16
3.5.2	Maintenance task listings validation and task selection.	15
3.11	Media selection.	29
3.12.7	Media selection and syllabus report.	39
3.14.3	Media selection and syllabus update report	40
3.11.1	Media selection model report	29
	Methodology	
3.7.1	Analysis of entry level behavior	18
3.21.3	External quality control planning.	65
3.21.1	Formative evaluation planning.	59
3.17.2.1	Hands-on lesson specification development.	50
3.23.1	Implementation plan.	68
3.24.2	Implementation plan update	71
3.21.2	Internal quality control planning.	62
3.19.1	Lesson formatting.	53
3.20.1	Lesson materials and tests	54
3.18.2	Lesson specification update.	53
3.5.1.2	Maintenance task listings development.	15
3.5.2.1	Maintenance task listings validation and task selection	15
3.8.2	Objectives hierarchies	19
3.9.2	Objectives hierarchies update.	25
3.6.2	Operator and maintenance task listings update.	17
3.4.1.2	Operator task listings development	7
3.4.2.1	Operator task listings validation and task selection	13
3.28.2.1	Program progress reporting	78
3.28.4.1	Program summary.	79
3.28.1.1	Program work plan.	77
3.28.3.1	Quality control during production and implementation	78
3.22.2	Quality control plan update.	68
3.10.4.1	Subsystem equipment commonality analysis (maintenance)	28
3.14.2	Syllabi update	40
3.2.1	Training development and support plan.	4
3.3.1	Training development and support plan update	7
3.15.1	Training support requirements analysis	40
3.16.1	Training support requirements analysis update.	43
3.1.1	Training system problem analysis	2
6.3	Miscellaneous notes.	82

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:25Z
Check the source to verify that this is the current version before use.

INDEX
(Continued)

Paragraph		Page
6	NOTES AND CONCLUDING MATERIAL.	81
3.8	Objectives hierarchies	18
3.8.4	Objectives hierarchies report.	24
3.9	Objectives hierarchies update.	24
3.9.3	Objectives hierarchies update report	25
3.6	Operator and maintenance task listings update.	17
3.10.3	Operator commonality analysis.	27
3.12	Operator syllabi	32
3.4	Operator task listings	7
3.4.1	Operator task listings development	7
3.4.3	Operator task listings report.	14
3.4.2	Operator task listings validation and task selection . . .	13
	Organization and sequencing	
3.13.2	Maintenance syllabi.	39
3.12.2	Operator syllabi	32
3.13.3	Other maintenance syllabi steps.	40
5	PACKAGING.	80
3.1.2	Problem analysis report.	4
3.21.5	Produce data forms and quality control manual.	65
3.28.2	Program progress reporting	78
3.28.4	Program summary.	79
3.28.1	Program work plan.	77
3.28.2.2	Progress reports	78
4	QUALITY ASSURANCE PROVISIONS	80
4.2	Quality assurance provisions, general.	80
3.28.3	Quality control during production and implementation . . .	78
3.21	Quality control plan	59
3.21.8	Quality control plan report.	67
3.22	Quality control plan update.	67
3.22.4	Quality control plan update report	68
3.28.3.2	Quality control report	79
4.4	Records.	80
6.3.1	Reference documents.	82
3	REQUIREMENTS	2
4.1	Responsibility for inspection.	80
6.3.2	Sample procedure for development of maintenance task listing.	82
1	SCOPE.	1
3.12.3	Selection of primary and alternate media for academic lessons.	37
3.10.1	Similar weapons system survey.	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T21:25Z
Check the source to verify that this is the current version before use.

INDEX
(Continued)

Paragraph		Page
3.10.5	State-of-the-art assessment.	28
3.7.3	Student entry level report	18
3.17.3	Student evaluation procedures.	51
3.27	Student orientation.	76
3.27.4	Student orientation manual	77
3.26.2	Student station(s)	75
3.17	Student training course lesson specifications.	43
3.20.4	Student training materials	59
	Subject matter expert training for	
3.27.2	Development of student orientation materials	77
3.26.4	Development of training device materials	75
3.25.2	Development of training system materials	74
3.7.2	Entry level behavior analysis.	18
3.20.3	Formative evaluation	58
3.23.2	Implementation plan development.	71
3.24.3	Implementation plan update	72
3.20.2	Lesson authoring	58
3.17.4	Lesson specification development	51
3.5.1.3	Maintenance task listings development.	15
3.5.2.2	Maintenance task listings validation and task selection	16
3.11.6	Media analysis	32
3.8.3	Objectives hierarchies development	24
3.4.1.3	Operator task listings development	12
3.4.2.2	Operator task listings validation and task selection	14
3.21.6	Quality control plan development	66
3.12.6	Syllabi development.	38
3.15.2	Training support requirements analysis	43
3.10.4	Subsystem equipment commonality analysis (maintenance)	27
3.14	Syllabi update	40
3.6.3	Task listings update report.	17
3.26.1	Trainer capabilities and characteristics	74
3.11.3	Trainer functional description report.	31
3.11.4	Trainer modification report.	31
3.26.3	Trainer operation.	75
3.2	Training development and support plan.	4
3.2.2	Training development and support plan report	7
3.3	Training development and support plan update	7
3.3.2	Training development and support plan update report.	7
3.26.6	Training device instructor/operator training materials	76
3.26	Training device operation.	74
3.10	Training equipment survey.	25
3.10.6	Training equipment survey report	28

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:25Z
Check the source to verify that this is the current version before use.

INDEX
(Continued)

Paragraph		Page
3.25.3	Training instructors in training system operation.	74
3.26.5	Training instructors/operators in training device operation.	75
3.20.5	Training materials development report.	59
3.23.3	Training of personnel in implementation activities	71
3.21.7	Training of personnel in quality control activities. . . .	66
3.24.4	Training of personnel in updated implementation activities	72
3.22.3	Training of personnel in updated quality control activities	68
3.28.4.2	Training program work report	80
3.15	Training support requirements analysis	40
3.16	Training support requirements analysis update.	43
3.16.2	Training support requirements analysis update report . . .	43
3.15.3	Training support requirements report	43
	Training system components, features, and operating procedures	
3.25.1	Instructor training.	72
3.27.1	Student orientation.	76
3.1	Training system problem analysis	2
3.28.1.2	Work plan report	78

Source: https://assist.dla.mil -- Downloaded: 202610-20T21:25Z
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